

Self-complementary tournament score sequences (OEIS A345470 and A351869)

Leading equivalents, the strong fraction $e^{-\lambda}$, the growth theorem,
monotonicity and threshold inverses

Two research reports of one external research session
(bundle Reports 116 and 114), merged into one ProveIt report

Parts I and II: 2 October 2026 merged: 5 October 2026

Abstract

A tournament score sequence $s_1 \leq \dots \leq s_n$ is self-complementary when $s_i + s_{n+1-i} = n - 1$. Let C_n count these sequences (OEIS A345470) and D_n the strong ones (A351869). This report prints two manuscripts as Parts I and II. Part I (Report 116, the base) proves $C_n \sim A 2^n n^{-3/4}$ and $D_n \sim e^{-\lambda} A 2^n n^{-3/4}$ with one amplitude $A > 0$ for both parities. Here $\lambda = 0.33023754\dots$ is the constant of the ordinary score sequences ($S(1/4) = e^\lambda$, Kolesnik), so the limiting strong fraction is $e^{-\lambda} = 0.71875297\dots$. Part I also proves a harmonic identity for the Denisov–Wachtel harmonic function, a Lambert- W smooth inverse with vanishing-width integer brackets, and an eventual zero-or-one gap between the strong and the all thresholds. Part II (Report 114) proved the bounds $\Theta(2^n n^{-3/4})$ first, by a bounded-bridge argument, with an $O(1)$ inverse, and shows that C_n is nondecreasing for all n and D_n for $n \geq 2$. Both manuscripts copy the limit density of the Denisov–Wachtel conditional limit theorem as printed (2015, p. 170). That display integrates to $0.1154552960\dots$, not 1. The write identifies the misprint: the display is not the density $\kappa^{-1} \tilde{h}(1, x, -y)$ that the same paper’s Remark 2 takes from Groeneboom, Jongbloed and Wellner; a factor s^{-2} and the constant are lost. The write proves that the corrected density has total mass exactly 1 and velocity marginal $\sqrt{2/3} \Gamma(3/4)^2 / \pi = 0.3902762\dots$ at 0, about ten times Part I’s printed value $0.0377237\dots$. Hence $A = \sqrt{6} \Gamma(3/4) \pi^{-3/2} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$, and the one proof step that used the misprinted shape is repaired. Exact counts suggest $A \approx 0.387$ (uncertified). A certified value of A , rates, correction terms and a direct proof of the harmonic identity remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 9c995cefe (batch 103):

Part	Bundle report	Delivered title	Labels
I	116 (base)	Leading equivalents for self complementary tournament score sequences	sct:
II	114	The growth of self complementary tournament score sequences	sct:gr:

Report 114 was written first; Report 116 cites it as “Report 114” for the Θ estimates and proves more. The placement made the stronger manuscript the base, so it is printed first. Part II keeps everything of Report 114, including what Part I proves again or more strongly (marked as second routes) and its monotonicity proposition, which Part I does not have.

Each Part prints its manuscript’s text, with every result, proof, remark, open question and limitation; statements keep their delivered wording. Sections and equations are numbered continuously through the report, and statements within sections, as in both deliveries. Part I keeps exactly the section, statement and equation numbers of Report 116 (Sections 1–12, equations (1)–(56)). In Part II, Report 114’s Section k is Section $k + 12$, its statement $k.j$ is $(k + 12).j$, and its equation (k) is equation $(k + 56)$. The write added Section 4.2 at the end of Section 4, Section 12.1 at the end of Section 12, and Section 21 at the end of Part II, so no delivered number moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date. They give cross-references between the Parts, mark results proved twice, record the correction of the limit density and connect the report to its neighbour **a000571-tournament-score-sequences**. Under Vladimir’s standing rule of 4 October 2026, every claim stated without proof is collected in Sections 12.1 and 21, with its source, its sketch and what is missing. The one claim found to be wrong stays printed as delivered, with its refutation in Remark 4.2.

The correction in brief. Both Parts use the limit law μ of Denisov–Wachtel’s Theorem 1 [1] and copy its displayed density, Part I with the printed constant (equation (18)), Part II with an unspecified constant $c_0 > 0$ (equation (76)). That display integrates to $0.1154552960\dots$, so it is not a probability density (Remark 4.2). Part I’s closed form (22) is therefore the value at 0 of the wrong function, and the amplitude formula (6) evaluated with it is about ten times too small; the exact counts confirm the factor. The density that Denisov–Wachtel’s own Remark 2 and proof refer to, $\bar{h}_\star = \kappa^{-1}\bar{h}_{\text{GJW}}(1, x, -y)$ with $\bar{h}_{\text{GJW}}$ from Groeneboom–Jongbloed–Wellner [2, (2.24)], has $s^{-5/2}$ where the display has $s^{-1/2}$ and the constant $2^{9/4}/\Gamma(1/4)$. Proposition 4.3 ([**write**]) proves that $\bar{h}_\star$ has mass exactly 1 and computes its velocity marginal at 0. Remark 4.4 shows that Lemma 4.1, the one step whose proof used the printed shape, holds for $\bar{h}_\star$ with a repaired majorant. Remark 4.5 lists what the misprint never touched. A dynamic programme for the conditioned walk to $n = 600$, made by the independent check of the write (dated paragraph at the end of Section 12.1), matches $\bar{h}_\star$ and rejects the printed display (Remark 4.5).

What is proved, and where

Here C_n and D_n count self-complementary and strong self-complementary score sequences (sorted sequences, not tournaments or isomorphism classes), $\lambda = \sum_k N_k/(k4^k)$, $r = e^{-\lambda}$, and V_{DW} is the Denisov–Wachtel harmonic function of the variance-one geometric-minus-one walk.

Statement	Where
$C_n, D_n = \Theta(2^n n^{-3/4})$; inverse	Part II, Theorem 13.1 (first proof); implied by
$N_E(x) = \log_2 x + \frac{3}{4} \log_2 \log_2 x + O(1)$	Part I
$C_n \sim A 2^n n^{-3/4}$, $D_n \sim r A 2^n n^{-3/4}$, one $A > 0$ for both parities	Part I, Theorem 1.1, with f_Y read as $f_\star$ (Remark 4.4)
$A = 2^{1/4} \kappa V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}) f_\star(0) = \sqrt{6} \Gamma(3/4) \pi^{-3/2} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$	(6) with [write] Proposition 4.3
$0.7187529767 < r < 0.7187529792$; $r = 1/S(1/4)$	Part I, Section 8
Exact encoding by an integrated geometric walk; the final slack is untested	Part I, Section 2; Part II, Proposition 14.1
$\mathbb{P}_z(\tau > n, V_n = b) \sim c_z f_\star(0) \sigma^{-1} n^{-3/4}$ for every positive start; $\Theta(k^{-3/4})$ from (1, 1)	Part I, Theorem 6.1; Part II, Lemma 16.1 (second route to the order)
Endpoint-weighted maximal estimate; bounded-bridge lower bound	Part I, Lemma 5.1; Part II, Lemma 15.2
Harmonic identity	Part I, (51)
$\sum_{g \geq 1} 2^{-g-1} V_{\text{DW}}(g/\sqrt{2}, g/\sqrt{2}) = e^{-\lambda} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$	
Lambert W_{-1} smooth inverse with constant term; vanishing-width ceiling brackets; $N_C \leq N_D \leq N_C + 1$ eventually, both gaps infinitely often	Part I, Proposition 10.1, Corollary 10.2
C_n nondecreasing for all $n \geq 0$, D_n for $n \geq 2$	Part II, Proposition 18.1 (not in Part I)
The corrected limit density $\bar{h}_\star$: total mass 1, $f_\star(0) = \sqrt{2/3} \Gamma(3/4)^2/\pi$; Lemma 4.1 for $\bar{h}_\star$	[write] Proposition 4.3, Remark 4.4
<i>Refuted</i>	
The display (18) (and (76)) as the density of μ ; the value 0.0377237... of (22) as the velocity marginal of μ at 0, and (6) evaluated with it: Remark 4.2.	
<i>Open in both Parts</i>	
A certified value of A (equivalently of $V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$); rates of convergence and correction terms, including the parity correction visible in the data; a direct proof of the harmonic identity; effective inverse brackets. See Sections 12.1 and 21.	

Finite computations in both Parts (exact counts, fixtures, mutation campaigns, the rational enclosure of λ) are checks of formulas and implementations; no asymptotic statement rests on them. The exceptions are the enclosures of λ and r , which combine exact partial sums with Kolesnik’s published tail bound.

Notation across the two Parts

Common to both Parts: $n = 2m$ or $2m + 1$, $M = \lfloor (n - 1)/2 \rfloor$; the gaps G_j with $\mathbb{P}(G_j = g) = 2^{-g-1}$ and $X_j = G_j - 1$; $\sigma = \sqrt{2}$; the velocity V_j , the area I_j of the walk started at (a, v) , the exit time τ ; the Gaussian transition density p_t of $(\int_0^t B, B_t)$ and $q_t = p_t(\cdot; u, v) - p_t(\cdot; u, -v)$; the Denisov–Wachtel cone K_+ , constant κ , harmonic function V_{DW} and limit law μ . No symbol of the sources was renamed. The symbols below change meaning between the Parts, inside one Part, or against the neighbouring report **a000571-tournament-score-sequences** (“a000571”).

Symbol	Meanings
C, D	Part I: counts C_n, D_n , series $C(z), D(z)$; C alone, C_z, C_0 are constants. Part II: counts $C(n), D(n)$; C_C, c_C, C_D, c_D, C_0 constants. Both: $C_0 = D_0 = 1$ by convention.
A	Part I: the amplitude A of Theorem 1.1, $A_C = A, A_D = rA$ (Section 10); also the Landau excess A_j (Section 2). Part II: the Landau excess A_r . a000571: $A(z)$ its logarithmic generating function, $A(1/4) = \lambda$. <i>False reading</i> : A is a constant, A_j a walk.
T	Part I: $T_j = X_1 + \dots + X_j$ (Sections 5–6), but $T_k, T(z)$ the strong <i>ordinary</i> score counts (Section 8), which are a000571's $I_k, I(z)$. Part II: T_j partial sums only.
I	Both Parts: I_j the area of the walk ($I_r = A_r + 1$ from the start $(1, 1)$). a000571: I_n the strong ordinary counts, which Part I calls T_n . <i>False reading</i> : Part I's I_j is not a000571's I_n .
S	Part I: $S_k, S(z)$ ordinary score counts (A000571), a000571's $S_n, S(z)$.
r	Part I: $r = e^{-\lambda}$; $r_n \in \{-1, 0\}$ the terminal velocity. Part II: r an index ($A_r, 1 \leq r \leq m$).
$\bar{h}, \bar{h}_*, c_{\text{DW}}, c_0, K_*$	$\bar{h}$: the density as printed in [1, Theorem 1] with $s^{-1/2}$, copied with constant c_{DW} (Part I) or $c_0 > 0$ (Part II); it is not a probability density (Remark 4.2). $\bar{h}_*$, $K_* = 2^{9/4}/\Gamma(1/4)$: the corrected density (Proposition 4.3). $\bar{h}_{\text{GJW}}(t, x, y)$: Groeneboom–Jongbloed–Wellner's function [2, (2.24)].
f_Y, F_a, f_*, F_a^*	Part I: $F_a, f_Y = F_0$ the velocity marginals of $\bar{h}$; Theorem 1.1 and Sections 6–7 need those of μ , which are F_a^*, f_* . <i>False reading</i> : $(22) = 0.0377\dots$ is not $f_*(0) = 0.3903\dots$
h	Part I: the slack index $h \geq 0$ in (13); Denisov–Wachtel's Brownian harmonic function h (Section 3); $h = \lfloor \ell/2 \rfloor$ in the proof of Lemma 5.1. Part II: $h = \lfloor \ell/2 \rfloor$ in Lemma 15.2.
q	Part I: q_t the Gaussian difference; $q_E(X)$ the smooth inverse (Section 10). Part II: q_u the Gaussian difference; $q = \ell - h$ in Lemma 15.2.
K	Part I: the velocity cutoff of (31), the convolution cutoff (Section 8), $K = 1669$ terms of the enclosure; K_+ the cone. Part II: the bridge constant $K > B$ of Lemma 15.2; K_+ .
N	Part I: N_k the Erdős–Ginzburg–Ziv numbers (4) (a000571's N_k , A145855); $N_E(X)$ thresholds with natural logarithms and real X . Part II: $N_E(x)$ with $\log_2$ and real x ; same thresholds.
a, b	Part I: a the starting area of $z = (a, v)$, and $a = 3/4$ in Section 10; b a terminal velocity. Part II: (a, b) the area–velocity arguments of the density (Part I's (x, y)), and b a terminal velocity in Lemma 16.1.
B	Part I: the direct strong amplitude (50), and the constant in $c_n \leq B(n+1)^{-3/4}$. Part II: the window constant of Lemmas 15.1–15.2.
c	Part I: $c_z = \kappa V_{\text{DW}}(a/\sigma, v/\sigma)$, $c_{(1,1)}$; $c_n = C_n/2^n$; $t_k = T_k/4^k$. Part II: c_* is Part I's $c_{(1,1)}$; c_B, c_L, c, c', c'' generic.
t, u, L	Part I: t the time in p_t , $t = \varepsilon/\alpha$ in Section 6; u a substitution; $L = \log X$. Part II: u the time in p_u ; $t = \lfloor k/2 \rfloor$ the halfway time; L an area level in Section 16.
μ, τ, σ	Both Parts: μ the limit law, τ the exit time, $\sigma = \sqrt{2}$. a000571: $\mu = \rho A'(\rho)$, τ a weight parameter, σ a sign.
λ	Both Parts and a000571: the same constant $\lambda = A(1/4) = \log S(1/4) = 0.3302375439859293\dots$

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243, both dated 2 October 2026, with the delivered author lines “Report 116” and “Report 114” (PDF metadata author “Research report”). Neither names an author or a tool, says it is AI-assisted, or carries “prepared for private review” wording, and neither pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by 9c995cfe (batch 103).

- **Part I**, bundle Report 116, archive `Self_Complementary_Tournament_Scores_Leading_Constants_and_Inverses_Source.zip` (541,891 bytes, 17 files; `report116.tex`, 900 lines, 14 pages). The base: its `report116.tex` became this `article.tex`. Its `checks/fixtures.json` pins Report 114's `checks/results.json` by SHA-256 (checked at placement). Contributes the leading equivalents with one amplitude, the strong fraction $e^{-\lambda}$ with its enclosure, the endpoint smoothing theorem, the harmonic identity, the inverses and threshold gap.
- **Part II**, bundle Report 114, archive `Self_Complementary_Tournament_Scores_Growth_and_Inverse_Source.zip` (483,412 bytes, 17 files; `report114.tex`, 717 lines, 11 pages). Contributes the first proof of the $\Theta(2^n n^{-3/4})$ bounds (Stockmeyer's Section 6 conjectures), the bounded-bridge lemma, the $O(1)$ inverse and the monotonicity proposition.

Where the merge had to choose:

1. *Base and order.* Report 116 is the base (stronger results) and is printed first, although Report 114 was written first. Report 116 credits Report 114 for the Θ estimates; the notes keep that priority.
2. *Union.* Every result, proof and limitation of Report 114 is printed, including the encoding, lemmas and inverse that Part I proves again or more strongly; they are marked as second routes, not removed.
3. *The density.* Both manuscripts copy the misprinted display of [1]. The delivered statements stay as printed. The write adds Section 4.2 (the refutation, the corrected density and the repaired lemma) and dated notes at every place that uses the display.
4. *Typography.* Labels were prefixed (`sct:`, `sct:gr:`) before anything cited them. Part II's seven `\nchoose2`-type binomials are typeset with `\binom` (same output, and the `amsmath` warning about `\atopwithdelims` disappears). The delivered title pages and running heads ("Report 116", "Report 114") are replaced by this report's; the letter paper, margins and theorem styles of both deliveries are kept. The two bibliographies are merged at the end (same keys, annotations of both kept), with three entries added in the write.
5. *Package text.* Section 11 and Section 20 describe the delivered packages under their delivery names; the README maps them to the shipped names and gives rerun routes.

Relation to the ordinary score sequences (a000571)

The collection's report `a000571-tournament-score-sequences` (same directory) treats ordinary score sequences: S_n (A000571), the strong ones I_n , $S(z) = e^{A(z)}$ and $I(z) = 1 - e^{-A(z)}$ with the singularity at $\rho = 1/4$. Its constant $\lambda = A(1/4) = 0.3302375439859293$ is exactly the λ of (3): Part I's T_k and S_k are its I_k and S_k , and (45) is its $S = e^A$. Its value lies inside Part I's certified enclosure $[0.3302375420437401, 0.3302375453489038]$, and $e^{-\lambda} = 0.71875297770\dots$ at its value lies inside both of Part I's intervals for r .

Why the strong fraction is $e^{-\lambda}$ here but $e^{-2\lambda}$ there. a000571 records Kolesnik's $S_n \sim e^\lambda 4^n n^{-5/2} / (2\sqrt{\pi})$ and $I_n \sim e^{-\lambda} 4^n n^{-5/2} / (2\sqrt{\pi})$, so $I_n/S_n \rightarrow e^{-2\lambda}$ (its equation `tss:eq:known`). Both are transforms of one singular series: if $a_n = [z^n]A(z) \sim 4^n n^{-5/2} / (2\sqrt{\pi})$, then $[z^n]e^{\pm A(z)} \sim \pm e^{\pm A(1/4)} a_n$, because $A(1/4)$ is finite and the singular part of $e^{\pm A}$ is $\pm e^{\pm A(1/4)}$ times that of A . So $S_n \sim e^\lambda a_n$ and $I_n = -[z^n]e^{-A(z)} \sim e^{-\lambda} a_n$ ($n \geq 1$): the all count gains one factor e^λ and the strong count loses one, and their ratio is $e^{-2\lambda}$. Here instead $D(z) = C(z)(1 - T(z^2)) = C(z)e^{-A(z^2)}$ by Stockmeyer's (39). The two factors are not transforms of one series: C_n is of order $2^n n^{-3/4}$, while the coefficients

of $e^{-A(z^2)}$ are of order $2^n n^{-5/2}$ (on even n) and summable at the radius $1/2$. The heavier factor dominates and the lighter one contributes only its value $e^{-A(1/4)} = 1/S(1/4) = e^{-\lambda}$, which is the convolution argument of Section 8. In particular $r = 1 - I(1/4)$ in a000571's notation.

Inverses. Part I's Lambert W_{-1} inverse (52) is the same mechanism as a000571's `tss:eq:lambert`, there for $4^x x^{-5/2}$ and here for $2^t t^{-3/4}$: a second instance, not a re-proof.

Other reports. No other report of the repository treats self-complementary score sequences, A345470, A351869, integrated random walks or the Denisov–Wachtel exit-time theorem (searched 5 October 2026); `a377922-corner-polyhedra-schnyder` and `a215561-fixed-composition-excursions` cite Denisov–Wachtel's paper on random walks in cones, a different paper. Nothing in this report is formalized in Lean or Rocq, and its placement confers no formal status.

Part I

Leading equivalents, the strong fraction and inverses

Bundle Report 116, archive `Self_Complementary_Tournament_Scores_Leading_Constants_and_Inverses_Source.zip`, delivered as `report116.tex` (14 pages), dated 2 October 2026. Delivered title “Leading equivalents for self complementary tournament score sequences”; author line “Report 116”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. “Report 114” in the text is Part II. Section 4.2 and Section 12.1 are added in the write.

Abstract of this Part (as delivered). Let C_n count self-complementary tournament score sequences of length n (OEIS A345470), and let D_n count the strong ones (A351869). We derive

$$C_n \sim A 2^n n^{-3/4}, \quad D_n \sim e^{-\lambda} A 2^n n^{-3/4},$$

with one positive amplitude A for both parities. The amplitude has an exact harmonic-function characterization; no numerical value for it is established. The limiting strong fraction satisfies $0.7187529762 < e^{-\lambda} < 0.7187529792$, using Kolesnik’s published bounds. The new probabilistic step upgrades a survival weak limit to a fixed-velocity endpoint equivalent by smoothing with a short final block. The proof treats marginal regularity, the killing error, final slack and convolution tails. It also gives a harmonic identity, a Lambert W smooth inverse, sharp vanishing-width ceiling brackets for integer thresholds, and an eventual zero-or-one threshold gap, with both values occurring infinitely often. No convergence rate or higher-order counting correction is proved.

`[write]` **5 October 2026.** The “exact harmonic-function characterization” of A is formula (6). It is correct when f_Y is the velocity marginal of the limit law μ . The “universal density factor” evaluated in Section 4 is computed from the density as printed in [1], which is misprinted, and its value is about ten times too small; see Section 4.2.

1 Result and conventions

A tournament score sequence is the sorted list of vertex outdegrees. Self-complementarity of a score sequence means

$$s_i + s_{n+1-i} = n - 1 \quad (1 \leq i \leq n). \quad (1)$$

We count distinct sorted sequences, not labeled tournaments or tournament isomorphism classes. A sequence is *strong* when its tournament realizations are strongly connected. Landau’s criterion and its strict version characterize these two classes by nonnegative and positive proper partial-sum excesses, respectively; see [5, Propositions 1 and 7].

Let C_n and D_n denote the all and strong self-complementary counts. Set $C_0 = D_0 = 1$ for the empty terms in generating functions. The small values are $C_1 = D_1 = 1$ and $C_2 = 1, D_2 = 0$; empty-term conventions do not assert that an empty tournament is strongly connected.

Theorem 1.1. *There exists $A > 0$ such that, through all positive integers n ,*

$$C_n \sim A 2^n n^{-3/4}, \quad D_n \sim r A 2^n n^{-3/4}, \quad r = e^{-\lambda}, \quad (2)$$

where

$$\lambda = \sum_{k \geq 1} \frac{N_k}{k4^k}, \quad (3)$$

$$N_k = \frac{1}{2k} \sum_{d|k} (-1)^{k+d} \varphi(k/d) \binom{2d}{d}. \quad (4)$$

Here φ is Euler's totient. In particular,

$$0.7187529762 < r < 0.7187529792. \quad (5)$$

For the variance-one geometric-minus-one walk defined in Section 3, with its Denisov–Wachtel normalization,

$$A = 2^{1/4} \kappa V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}) f_Y(0) > 0. \quad (6)$$

Thus the same leading constants hold on even and odd lengths.

[write] **5 October 2026, reading of f_Y .** In (6), $f_Y(0)$ must be the density at 0 of the velocity marginal of the limit law μ of (17). Section 4 defines f_Y from the density displayed in [1, Theorem 1], which has total mass 0.1154552960..., not 1 (Remark 4.2). With f_Y read as the marginal $f_\star$ of the corrected density $\bar{h}_\star$ of Proposition 4.3, the theorem and its proof stand (Remark 4.4), and

$$A = 2^{1/4} \kappa V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}) f_\star(0) = \frac{\sqrt{6} \Gamma(3/4)}{\pi^{3/2}} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}) = 0.53905690 \dots \times V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}),$$

by $f_\star(0) = \sqrt{2/3} \Gamma(3/4)^2/\pi$, (15) and $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$. With the printed value (22) the formula would give an A about ten times too small. The two equivalents, their common amplitude and $r = e^{-\lambda}$ do not depend on the value of $f_Y(0)$ (Remark 4.5).

Report 114 established the two $\Theta(2^n n^{-3/4})$ estimates, resolving the order-of-growth questions in [5, Section 6]. This separate report strengthens that conclusion and recalls the encoding so that its proof can be read independently. The underlying ordinary-score theorems, generating functions and numerical bound on λ are published results and are attributed below. The application here concerns the self-complementary counts. No claim of publication, peer review or priority is made.

[write] **5 October 2026.** Report 114 is Part II of this report: its Theorem 13.1 is the Θ estimate with the inverse (59). Part II also proves, in Proposition 18.1, that C_n is nondecreasing for all n and D_n for $n \geq 2$; this Part has only eventual strict increase (Section 10).

1.1 Proof outline and scope

An exact geometric-gap encoding gives $2^{-n} C_n$ as a survival probability with a final velocity constraint. Denisov–Wachtel Theorem 1 supplies survival of order $n^{-1/4}$ and a conditional weak limit. A free last block supplies one-dimensional endpoint mass of order $n^{-1/2}$. The substantive step is to obtain its *constant* while controlling the paths killed in that block. Bounded continuity of the limiting velocity marginal and an endpoint-weighted maximal estimate make this possible. A summable slack mixture then proves the first equivalent. Stockmeyer's exact convolution and the ordinary-score tail bound give the second. The proof uses no killed local-limit formula, and specifically does not use equation (12) of [1].

2 Exact half-score and slack encoding

Write $n = 2m$ or $n = 2m + 1$ and

$$M = \left\lfloor \frac{n-1}{2} \right\rfloor = \begin{cases} m-1, & n = 2m, \\ m, & n = 2m+1. \end{cases}$$

For $m \geq 1$, every self-complementary candidate is uniquely determined by

$$0 \leq s_1 \leq \dots \leq s_m \leq M. \quad (7)$$

Reflect by (1), inserting $s_{m+1} = m$ in the odd case. The result is nondecreasing and its total is $\binom{n}{2}$. For the full sequence put

$$A_j = \sum_{i=1}^j s_i - \binom{j}{2} \quad (0 \leq j \leq n).$$

The last j scores sum to $j(n-1) - \sum_{i \leq j} s_i$, and therefore

$$A_{n-j} = \binom{n}{2} - j(n-1) + \sum_{i \leq j} s_i - \binom{n-j}{2} = A_j.$$

Consequently all Landau inequalities reduce exactly to

$$A_j \geq 0 \quad (1 \leq j \leq m); \quad \text{strongness replaces } \geq 0 \text{ by } > 0. \quad (8)$$

This includes the middle inequality, and both middle inequalities for odd n .

Define the gaps and the *last slack*

$$G_1 = s_1, \quad G_j = s_j - s_{j-1} \quad (2 \leq j \leq m), \quad G_{m+1} = M - s_m. \quad (9)$$

This is a bijection between (7) and nonnegative compositions $\sum_{j=1}^{m+1} G_j = M$. Take the G_j independent with

$$\mathbb{P}(G_j = g) = 2^{-g-1}, \quad g \in \mathbb{Z}_{\geq 0}, \quad X_j = G_j - 1. \quad (10)$$

Every such composition has probability $2^{-M-m-1} = 2^{-n}$. With initial velocity $V_0 = 1$ and area $A_0 = 0$, set

$$V_j = 1 + \sum_{i=1}^j X_i, \quad A_j = \sum_{i=1}^j V_i.$$

For $j \leq m$, $V_j = s_j - j + 1$, so A_j is precisely the Landau excess. The composition condition is equivalent to

$$V_{m+1} = r_n, \quad r_n = \begin{cases} -1, & n = 2m, \\ 0, & n = 2m+1. \end{cases} \quad (11)$$

For a general state $z = (a, v)$ we henceforth use

$$V_j = v + \sum_{i=1}^j X_i, \quad I_j = a + \sum_{i=1}^j V_i, \quad \tau = \min\{j \geq 1 : I_j \leq 0\}. \quad (12)$$

Because A_j is integer, $A_j \geq 0$ is equivalent to $I_j = A_j + 1 > 0$. Thus weak inequalities correspond to the start $(I_0, V_0) = (1, 1)$. Conditioning on the independent last slack $G_{m+1} = h$ gives the exact formula

$$2^{-n}C_n = \sum_{h \geq 0} 2^{-h-1} \mathbb{P}_{(1,1)}(\tau > m, V_m = r_n + 1 - h). \quad (13)$$

The area at time $m + 1$ is deliberately untested. For $n = 2$, the valid weak half sequence (0) would fail a new test on its final unshifted area; for $n = 4$, the strong half sequence (1, 1) has final unshifted area zero and would fail a strict extra test. The slack is not another score.

[write] 5 October 2026. Part II, Proposition 14.1, proves the same encoding, with the same two slack counterexamples; (13) refines it by conditioning on the last slack. Part II writes the excess A_r and the endpoint as V_{m+1} ; here the endpoint is moved to time m by the slack mixture.

3 The imported survival and weak-limit theorem

For (10),

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = \sigma^2 = 2, \quad \mathbb{E}e^{\theta|X|} < \infty \quad (0 < \theta < \log 2). \quad (14)$$

Its support $\{-1, 0, 1, \dots\}$ has maximal lattice span one. Apply Denisov–Wachtel [1, Theorem 1, (7)–(8), pp. 169–170] to X/σ . Its assumptions are mean zero, variance one and a finite $(2+\delta)$ -moment for some $\delta > 0$, all verified by (14). The theorem’s positive harmonic function V_{DW} is normalized as in that paper; its universal survival constant is

$$\kappa = \frac{3\Gamma(1/4)}{2^{3/4}\pi^{3/2}}. \quad (15)$$

For a fixed point in

$$K_+ = \{z \in (0, \infty) \times \mathbb{R} : \mathbb{P}_z(Z_j \in (0, \infty)^2, \tau > j) > 0 \text{ for some } j \geq 0\},$$

the theorem provides the survival equivalent and the conditional endpoint weak limit. In this definition Z_j denotes the area–velocity pair of the variance-one process. Every positive-coordinate start belongs to K_+ at $j = 0$.

Divide *both* coordinates of (12) by σ . For any fixed positive-coordinate integer state $z = (a, v)$ this gives

$$\mathbb{P}_z(\tau > n) \sim c_z n^{-1/4}, \quad c_z = \kappa V_{\text{DW}}(a/\sigma, v/\sigma) > 0, \quad (16)$$

$$\mathcal{L}\left(\frac{I_n}{\sigma n^{3/2}}, \frac{V_n}{\sigma \sqrt{n}} \mid \tau > n\right) \Rightarrow \mu. \quad (17)$$

The measure μ is universal for these standardized walks. Theorem 1 has no lattice condition; the span-one calculations below remain in the original integer coordinates. In particular, the endpoint mass will contribute $1/\sigma$, not a two-dimensional lattice-cell factor.

For precision, define V_{DW} using the paper’s normalization $V_{\text{DW}}(z) = \lim_{n \rightarrow \infty} \mathbb{E}_z[h(Z_n); \tau > n]$, where h is its Brownian harmonic function [1, equation (4)]. This fixes the scale of (6); it is not an arbitrary positive harmonic function. Equivalently, the product $\kappa V_{\text{DW}}(z)$ is fixed by (16).

[write] 5 October 2026. The density of μ is displayed in [1, Theorem 1] with a misprint, and Section 4 copies that display. The paper’s own Remark 2 identifies the density as $\kappa^{-1} \bar{h}_{\text{GJW}}(1, x, -y)$ with $\bar{h}_{\text{GJW}}$ from [2, (2.24)]; this is the corrected density $\bar{h}_*$ of Section 4.2. The survival equivalent (16), the weak limit (17) and the normalization of V_{DW} are not affected.

4 Regularity of the limiting velocity density

We must localize at velocity zero. Abstract weak convergence alone does not supply this step. We verify the required regularity from the density displayed in [1, Theorem 1]. Write

$$\begin{aligned}\bar{h}(x, y) &= c_{\text{DW}} \int_0^1 \int_0^\infty w^{3/2} s^{-1/2} e^{-2w^2/s} q_{1-s}(x, -y; 0, -w) dw ds, \\ c_{\text{DW}} &= \frac{2^{9/4} \sqrt{\pi}}{\sqrt{3} \Gamma(1/4)}, \quad q_t(x, y; u, v) = p_t(x, y; u, v) - p_t(x, y; u, -v),\end{aligned}\tag{18}$$

where the free Gaussian density is

$$p_t(x, y; u, v) = \frac{\sqrt{3}}{\pi t^2} \exp \left(-\frac{6(u-x-ty)^2}{t^3} + \frac{6(u-x-ty)(v-y)}{t^2} - \frac{2(v-y)^2}{t} \right).\tag{19}$$

For $a \geq 0$ define

$$F_a(y) = \int_{x>a} \bar{h}(x, y) dx, \quad f_Y(y) = F_0(y).\tag{20}$$

[write] **5 October 2026, correction.** The display (18) copies [1, p. 170] exactly, and that display is misprinted: it integrates to $0.1154552960\dots$, so it is not the density of μ (Remark 4.2). The density of μ is $\bar{h}_*$ of Proposition 4.3: $s^{-1/2}$ must read $s^{-5/2}$, and c_{DW} must read $K_* = 2^{9/4}/\Gamma(1/4)$. The proof of Lemma 4.1 below is correct for the display as printed. For $\bar{h}_*$ the lemma remains true, but its majorant must be replaced; see Remark 4.4.

Lemma 4.1. *The functions $F_a(y)$ are jointly continuous in $(a, y) \in [0, \infty) \times \mathbb{R}$ and bounded by a single finite constant. Moreover $0 < f_Y(0) < \infty$.*

Proof. Completing the square in the starting area x gives

$$\int_{\mathbb{R}} p_t(x, -y; 0, \pm w) dx = (2\pi t)^{-1/2} \exp \left(-\frac{(y \pm w)^2}{2t} \right).\tag{21}$$

Consequently

$$\int_{x>a} |q_t(x, -y; 0, -w)| dx \leq 2(2\pi t)^{-1/2}.$$

Multiplying by the weight in (18) and integrating over w yields a constant times

$$s^{-1/2}(1-s)^{-1/2} \int_0^\infty w^{3/2} e^{-2w^2/s} dw = C s^{3/4} (1-s)^{-1/2}.$$

This is integrable over $0 < s < 1$. Thus Fubini's theorem applies absolutely to the integrals defining F_a , and they share a bound independent of a, y . For fixed s, w , the integral over $x > a$ of either Gaussian in (21) is a continuous Gaussian-tail function of a, y . The same integrable majorant and dominated convergence prove joint continuity, including at $a = 0$.

Finally, for $x, w, t > 0$,

$$\frac{p_t(x, 0; 0, -w)}{p_t(x, 0; 0, w)} = \exp(12wx/t^2) > 1.$$

All outer weights in (18) are positive, so $\bar{h}(x, 0) > 0$ for $x > 0$. Hence $F_a(0) > 0$ for every finite a , and the claimed assertion at zero follows. $\square$

This argument selects continuous representatives of the marginal densities; it does not rely on their almost-everywhere values at one chosen point.

4.1 An explicit universal factor

The factor $f_Y(0)$ admits a convergent special-function expression independent of the discrete harmonic function:

$$f_Y(0) = \frac{5\Gamma(7/4)^2}{96\sqrt{2\pi}} {}_3F_2 \left(\begin{matrix} 1, 7/4, 9/4 \\ 3/2, 4 \end{matrix}; \frac{3}{4} \right). \quad (22)$$

Here ${}_pF_q$ has its usual series definition $\sum_{j \geq 0} (a_1)_j \cdots (a_p)_j z^j / ((b_1)_j \cdots (b_q)_j j!)$, where $(a)_j$ is the rising factorial. The series in (22) is positive and convergent.

To derive the formula, integration of the two completed Gaussian squares over $x > 0$ gives, with $u = 1 - s$,

$$\int_0^\infty q_u(x, 0; 0, -w) dx = \frac{e^{-w^2/(2u)}}{\sqrt{2\pi u}} \operatorname{erf} \left(\sqrt{\frac{3}{2u}} w \right).$$

For $P, Q > 0$, the resulting w integral is

$$\int_0^\infty w^{3/2} e^{-Pw^2} \operatorname{erf}(Qw) dw = \frac{Q\Gamma(7/4)}{\sqrt{\pi} P^{7/4}} {}_2F_1 \left(\begin{matrix} 1, 7/4 \\ 3/2 \end{matrix}; -\frac{Q^2}{P} \right). \quad (23)$$

For example, insert $\operatorname{erf}(Qw) = 2Qw\pi^{-1/2} \int_0^1 e^{-Q^2 w^2 t^2} dt$ and first evaluate the nonnegative gamma integral in w . Substitute $P = (4 - 3s)/(2s(1 - s))$, $Q = \sqrt{3/(2(1 - s))}$. Pfaff's transformation changes the hypergeometric factor to

$$\left(\frac{4 - 3s}{4} \right)^{7/4} {}_2F_1 \left(\begin{matrix} 1, 7/4 \\ 3/2 \end{matrix}; \frac{3s}{4} \right).$$

After cancellation of powers, (18) therefore gives

$$f_Y(0) = \frac{\Gamma(7/4)}{\sqrt{2\pi}\Gamma(1/4)} \int_0^1 s^{5/4} (1 - s)^{3/4} {}_2F_1 \left(\begin{matrix} 1, 7/4 \\ 3/2 \end{matrix}; \frac{3s}{4} \right) ds.$$

Termwise integration is justified by positivity and convergence at $3/4$. The beta integral contributes $B(9/4, 7/4)$ and the additional hypergeometric parameters $9/4$ and 4 . Finally $\Gamma(9/4) = 5\Gamma(1/4)/16$ and $\Gamma(4) = 6$ give (22). This exact evaluation of the universal factor does not evaluate A , which still contains $V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$.

[write] 5 October 2026, correction. This evaluation is correct for the printed display: its value is $0.0377237238918081803966\dots$, and a quadrature of (18) at the placement agreed to 30 digits. It is not the velocity marginal of μ at 0. The same steps with $s^{-5/2}$ in place of $s^{-1/2}$ give $f_\star(0) = \sqrt{2/3}\Gamma(3/4)^2/\pi = 0.3902762188691763\dots$, which is $10.3456\dots$ times larger (Proposition 4.3(b)). The ratio is not the factor 8.6614 by which the mass is off, because the misprint changes the shape as well as the constant.

4.2 The limit density: a misprint and its correction

[write] (5 October 2026.) This subsection was added in the write. It records a claim, shared by both Parts, that is demonstrably wrong (under Vladimir's standing rule of 4 October 2026), identifies its source and corrects it. Throughout, $u = 1 - s$, and for $v \in \mathbb{R}$

$$G_a(v) = \int_{x>a} p_u(x, -y; 0, v) dx, \quad \int_{x>a} q_u(x, -y; 0, -w) dx = G_a(-w) - G_a(w).$$

Remark 4.2 ([write], wrong claim). The display (18), copied from [1, Theorem 1, p. 170] by this Part and, with an unspecified constant $c_0 > 0$, by Part II as (76), is not a probability density. Its total mass is

$$\int_0^\infty \int_{\mathbb{R}} \bar{h}(x, y) dy dx = 0.11545529600527263025801512135 \dots$$

Consequently the value $0.0377237 \dots$ of (22) is not the density at 0 of the velocity marginal of μ , and (6) evaluated with it does not give A . The statements are kept as printed.

Proof of the counterexample. Write $d = uy - x$ and $m = u(v + y)/2$. Completing the square in d in (19) gives

$$p_u(x, -y; 0, v) = \frac{\sqrt{3}}{\pi u^2} \exp\left(-\frac{(v + y)^2}{2u} - \frac{6(d - m)^2}{u^3}\right),$$

and integrating over $x > a$, that is over $d < uy - a$,

$$G_a(v) = \frac{e^{-(v+y)^2/(2u)}}{2\sqrt{2\pi u}} \left(1 + \operatorname{erf}\left(\sqrt{6} \frac{u(y - v)/2 - a}{u^{3/2}}\right)\right).$$

For $a = 0$ the factor $e^{-(v+y)^2/(2u)}/\sqrt{2\pi u}$ is the density in y of $N(-v, u)$, under which $y - v \sim N(-2v, u)$. The Gaussian identity $\mathbb{E} \operatorname{erf}(cZ) = \operatorname{erf}(c\mathbb{E}Z/\sqrt{1 + 2c^2 \operatorname{Var} Z})$ with $c = \sqrt{3/(2u)}$, so that $1 + 2c^2u = 4$, gives $\int_{\mathbb{R}} G_0(v) dy = \frac{1}{2}(1 - \operatorname{erf}(v\sqrt{3/(2u)}))$, hence

$$\int_{\mathbb{R}} \int_{x>0} q_u(x, -y; 0, -w) dx dy = \operatorname{erf}(w\sqrt{3/(2u)}).$$

The integrand of (18) is absolutely integrable over (s, w, x, y) by the majorant in the proof of Lemma 4.1 (with the bound $\int \int |q_u| dx dy \leq 2$). By Fubini the mass is therefore

$$c_{\text{DW}} \int_0^1 s^{-1/2} \int_0^\infty w^{3/2} e^{-2w^2/s} \operatorname{erf}\left(w\sqrt{\frac{3}{2(1-s)}}\right) dw ds,$$

whose inner integral is (23) with $P = 2/s$ and $Q^2 = 3/(2(1 - s))$. Quadrature gives the value above (40 digits with mpmath at the placement; 12 digits by an independent scipy quadrature in the write). It would have to be 1.

Counts against the prediction. By Section 7, $A = 2^{3/4} c_{(1,1)} f_Y(0)/\sigma = 2^{1/4} c_{(1,1)} f_Y(0)$ with $c_{(1,1)} = \lim n^{1/4} \mathbb{P}_{(1,1)}(\tau > n)$, which neither Part evaluates. A dynamic programme for the geometric-minus-one walk, made by the independent check of the write (floating point on a truncated state space, discarded mass below 10^{-9}), gives $n^{1/4} \mathbb{P}_{(1,1)}(\tau > n) = 0.839569, 0.837671, 0.836094, 0.835866$ at $n = 100, 200, 500, 600$, extrapolating to $c_{(1,1)} \approx 0.83360$. With (22) the formula then predicts $A \approx 0.0374$. The exact counts (floating-point dynamic programming at the placement) give $C_n n^{3/4} 2^{-n} = 0.3606$ and 0.4249 at $n = 40, 41$, and 0.3762 and 0.3989 at $n = 320, 321$; the even values increase, the odd values decrease, with a gap of order $n^{-1/2}$, and the parity average extrapolates to $A \approx 0.387$. Both extrapolations are uncertified, but the factor of ten is far outside their uncertainty. (Dated note, 5 October 2026, after the independent check of the write: the second and third sentences of this paragraph first read “A Monte Carlo of the geometric-minus-one walk at the placement (480,000 paths each) gave $n^{1/4} \mathbb{P}_{(1,1)}(\tau > n) = 0.8355$ at $n = 500$ and 0.8426 at $n = 2000$. With (22) the formula then predicts $A \approx 0.038$.” The value 0.038 over-rounds: $2^{1/4} \cdot 0.83360 \cdot 0.0377237 = 0.03740$, and the two Monte Carlo values give 0.0375 and 0.0378 . The Monte Carlo value 0.8355 at $n = 500$ agrees with the dynamic programme’s 0.836094 ; 0.8426 at $n = 2000$ lies about 0.009 above its values, which decrease monotonically towards 0.83360 .)

Proposition 4.3 ([write]). *Let $K_\star = 2^{9/4}/\Gamma(1/4)$ and, for $x > 0$ and $y \in \mathbb{R}$,*

$$\bar{h}_\star(x, y) = K_\star \int_0^1 \int_0^\infty w^{3/2} s^{-5/2} e^{-2w^2/s} q_{1-s}(x, -y; 0, -w) dw ds.$$

Then $\bar{h}_\star(x, y) = \kappa^{-1} \bar{h}_{\text{GJW}}(1, x, -y)$, where $\bar{h}_{\text{GJW}}$ is the function [2, (2.24)] named in [1, Remark 2]. The integral converges absolutely for almost every (x, y) , and:

- (a) $\int_0^\infty \int_{\mathbb{R}} \bar{h}_\star(x, y) dy dx = 1$;
- (b) the velocity marginal $f_\star(y) = \int_{x>0} \bar{h}_\star(x, y) dx$ has the value

$$f_\star(0) = \sqrt{2/3} \Gamma(3/4)^2 / \pi = 0.3902762188691763 \dots;$$

- (c) the integrand of $\bar{h}_\star(x, y)$ is positive for $x > 0$ and $y \leq 3x$; in particular $\int_{x>a} \bar{h}_\star(x, 0) dx > 0$ for every $a \geq 0$.

The display (18) differs from $\bar{h}_\star$ in the power of s ($s^{-1/2}$ for $s^{-5/2}$: the factor s^{-2} of $p_s(0, w; 0, 0)$ is lost) and in the constant ($c_{\text{DW}} = \sqrt{\pi/3} K_\star$).

Proof. Identification. By (19), $p_s(0, w; 0, 0) = \sqrt{3} e^{-2w^2/s} / (\pi s^2)$. Groeneboom, Jongbloed and Wellner define

$$\bar{h}_{\text{GJW}}(t, x, y) = \frac{4\sqrt{3}}{\sqrt{2\pi}} \int_0^t \int_0^\infty w^{3/2} s^{-1/2} p_s(0, w; 0, 0) q_{t-s}(x, y; 0, -w) dw ds$$

[2, (2.24)] (the same formula is restated in [3, Section 6]). At $t = 1$, with y replaced by $-y$, this is $\frac{12}{\pi\sqrt{2\pi}} \int_0^1 \int_0^\infty w^{3/2} s^{-5/2} e^{-2w^2/s} q_{1-s}(x, -y; 0, -w) dw ds$, and $\kappa^{-1} \cdot 12/(\pi\sqrt{2\pi}) = 2^{9/4}/\Gamma(1/4)$ by (15).

Two bounds on q . The formula for G_a in Remark 4.2 gives $0 \leq G_a(v) \leq (2\pi u)^{-1/2}$, and $\int_{\mathbb{R}} \int_{\mathbb{R}} p_u dx dy = 1$. Moreover $\partial_v \log p_u(x, -y; 0, v) = 6(d-m)/u^2 - (v+y)/u$, and $\int_{\mathbb{R}} e^{-6z^2/u^3} dz = (\pi u^3/6)^{1/2}$, $\int_{\mathbb{R}} |z| e^{-6z^2/u^3} dz = u^3/6$, $\sup_z |z| e^{-z^2/(2u)} = (u/e)^{1/2}$. Hence

$$\int_{\mathbb{R}} |\partial_v p_u(x, -y; 0, v)| dx \leq \frac{3}{u}, \quad \int_{\mathbb{R}} \int_{\mathbb{R}} |\partial_v p_u(x, -y; 0, v)| dx dy \leq \frac{3}{\sqrt{u}}.$$

Since $q_u(x, -y; 0, -w) = -\int_{-w}^w \partial_v p_u(x, -y; 0, v) dv$, for all $a \geq 0$, $y \in \mathbb{R}$, $w > 0$ and $u \in (0, 1)$,

$$\int_{x>a} |q_u(x, -y; 0, -w)| dx \leq \min\left\{\frac{2}{\sqrt{2\pi u}}, \frac{6w}{u}\right\}, \quad \int_{\mathbb{R}} \int_{x>0} |q_u(x, -y; 0, -w)| dx dy \leq \min\left\{2, \frac{6w}{\sqrt{u}}\right\}.$$

An integrable majorant. Put $W(s, w) = w^{3/2} s^{-5/2} e^{-2w^2/s}$ and

$$M(s, w) = W(s, w) \left(\frac{6w}{1-s} \mathbf{1}_{\{s \leq 1/2\}} + \frac{2}{\sqrt{2\pi(1-s)}} \mathbf{1}_{\{s > 1/2\}} \right).$$

For $s \leq 1/2$, $\int_0^\infty M dw \leq 12s^{-5/2} \int_0^\infty w^{5/2} e^{-2w^2/s} dw = 12 \cdot 2^{-11/4} \Gamma(7/4) s^{-3/4}$. For $s > 1/2$, $\int_0^\infty M dw \leq 2^{7/2} (2\pi(1-s))^{-1/2} \int_0^\infty w^{3/2} e^{-2w^2} dw$. Both are integrable, so $\int_0^1 \int_0^\infty M < \infty$, and M dominates $W(s, w) \int_{x>a} |q_{1-s}(x, -y; 0, -w)| dx$ for every $a \geq 0$ and y . In the same way $W(s, w) \min\{2, 6w(1-s)^{-1/2}\}$, which dominates $W(s, w) \int_{\mathbb{R}} \int_{x>0} |q_{1-s}| dx dy$, is integrable. By Tonelli the integral defining $\bar{h}_\star$ converges absolutely for almost every (x, y) , and every integral of $\bar{h}_\star$ over $x > a$, or over $x > 0$ and $y \in \mathbb{R}$, may be computed in any order.

(a) By the identity in Remark 4.2, the mass is $K_\star \int_0^1 s^{-5/2} J(s) ds$, where $Q = (3/(2(1-s)))^{1/2}$ and

$$J(s) = \int_0^\infty w^{3/2} e^{-2w^2/s} \operatorname{erf}(Qw) dw, \quad \operatorname{erf}(Qw) = \frac{2Qw}{\sqrt{\pi}} \int_0^1 e^{-Q^2 w^2 t^2} dt.$$

Inserting the second formula in the first and integrating in w (all terms positive),

$$s^{-5/2} J(s) = \frac{\Gamma(7/4) Q}{\sqrt{\pi}} \int_0^1 \frac{s^{-5/2} dt}{(2/s + Q^2 t^2)^{7/4}} = \frac{\Gamma(7/4)}{2^{7/4} \sqrt{\pi}} \sqrt{\frac{3}{2}} \int_0^1 \frac{s^{-3/4} (1-s)^{5/4}}{(1-s(1-3t^2/4))^{7/4}} dt.$$

For fixed $t \in (0, 1]$ put $z = 1 - 3t^2/4 \in [1/4, 1)$. Euler's integral $\int_0^1 s^{b-1} (1-s)^{c-b-1} (1-zs)^{-a} ds = B(b, c-b) F(a, b; c; z)$ with $(a, b, c) = (7/4, 1/4, 5/2)$, followed by Euler's transformation $F(a, b; c; z) = (1-z)^{c-a-b} F(c-a, c-b; c; z)$, gives

$$\int_0^1 \frac{s^{-3/4} (1-s)^{5/4}}{(1-zs)^{7/4}} ds = B(\frac{1}{4}, \frac{9}{4}) \frac{\sqrt{3}t}{2} F(\frac{3}{4}, \frac{9}{4}; \frac{5}{2}; z).$$

Since $\frac{d}{dz} F(-\frac{1}{4}, \frac{5}{4}; \frac{3}{2}; z) = -\frac{5}{24} F(\frac{3}{4}, \frac{9}{4}; \frac{5}{2}; z)$ and $t dt = -\frac{2}{3} dz$,

$$\int_0^1 \frac{\sqrt{3}t}{2} F(\frac{3}{4}, \frac{9}{4}; \frac{5}{2}; 1 - \frac{3}{4}t^2) dt = \frac{1}{\sqrt{3}} \int_{1/4}^1 F(\frac{3}{4}, \frac{9}{4}; \frac{5}{2}; z) dz = \frac{24}{5\sqrt{3}} \left[F(-\frac{1}{4}, \frac{5}{4}; \frac{3}{2}; \frac{1}{4}) - F(-\frac{1}{4}, \frac{5}{4}; \frac{3}{2}; 1) \right];$$

the integrand has an integrable singularity $(1-z)^{-1/2}$ at $z = 1$, and $F(-\frac{1}{4}, \frac{5}{4}; \frac{3}{2}; \cdot)$ is continuous on $[0, 1]$ because $\frac{3}{2} + \frac{1}{4} - \frac{5}{4} > 0$. The quadratic case $F(\alpha, 1-\alpha; \frac{3}{2}; \sin^2 \theta) = \sin((2\alpha-1)\theta)/((2\alpha-1)\sin \theta)$ [4, §15.4(i)] with $\alpha = -\frac{1}{4}$ gives $2\sqrt{2}/3$ at $\theta = \pi/6$ and $\sqrt{2}/3$ at $\theta = \pi/2$ (Gauss's sum gives the same). So the t -integral is $8\sqrt{2}/(5\sqrt{3})$. With $B(\frac{1}{4}, \frac{9}{4}) = \Gamma(\frac{1}{4})\Gamma(\frac{9}{4})/\Gamma(\frac{5}{2}) = 5\Gamma(1/4)^2/(12\sqrt{\pi})$,

$$\iint \bar{h}_\star = K_\star \frac{\Gamma(7/4)}{2^{7/4} \sqrt{\pi}} \sqrt{\frac{3}{2}} \cdot \frac{5\Gamma(1/4)^2}{12\sqrt{\pi}} \cdot \frac{8\sqrt{2}}{5\sqrt{3}} = K_\star \frac{2\Gamma(7/4)\Gamma(1/4)^2}{3 \cdot 2^{7/4} \pi} = K_\star 2^{-9/4} \Gamma(1/4) = 1,$$

using $\Gamma(7/4) = \frac{3}{4}\Gamma(\frac{3}{4})$ and $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$.

(b) For $a = y = 0$ the formula for G_0 gives the inner integral of Section 4,

$$G_0(-w) - G_0(w) = (2\pi u)^{-1/2} e^{-w^2/(2u)} \operatorname{erf}(w\sqrt{3/(2u)}).$$

The steps of that section, with $s^{-5/2}$ in place of $s^{-1/2}$ (so with (23) for $P = (4-3s)/(2s(1-s))$ and Pfaff's transformation), give

$$f_\star(0) = K_\star \frac{\sqrt{3}\Gamma(7/4)}{2^{11/4}\pi} \int_0^1 s^{-3/4} (1-s)^{3/4} F(1, \frac{7}{4}; \frac{3}{2}; \frac{3s}{4}) ds = K_\star \frac{\sqrt{3}\Gamma(7/4)}{2^{11/4}\pi} B(\frac{1}{4}, \frac{7}{4}) {}_3F_2\left(\frac{1}{4}, 1, \frac{7}{4}; \frac{3}{2}, 2; \frac{3}{4}\right),$$

by termwise integration of positive terms. Because two parameters are 1 and 2, ${}_3F_2(a, 1, b; c, 2; z) = \frac{c-1}{(a-1)(b-1)z} (F(a-1, b-1; c-1; z) - 1)$, and $F(-\frac{3}{4}, \frac{3}{4}; \frac{1}{2}; \sin^2 \theta) = \cos(\frac{3}{2}\theta)$ [4, §15.4(i)] vanishes at $\sin^2 \theta = \frac{3}{4}$. Hence the ${}_3F_2$ equals $-\frac{32}{27}(0-1) = \frac{32}{27}$, and with $B(\frac{1}{4}, \frac{7}{4}) = \Gamma(\frac{1}{4})\Gamma(\frac{7}{4})$,

$$f_\star(0) = \frac{2^{9/4}}{\Gamma(1/4)} \cdot \frac{\sqrt{3}\Gamma(7/4)^2\Gamma(1/4)}{2^{11/4}\pi} \cdot \frac{32}{27} = \sqrt{\frac{3}{2}} \cdot \frac{9}{16} \cdot \frac{32}{27} \cdot \frac{\Gamma(3/4)^2}{\pi} = \sqrt{\frac{2}{3}} \frac{\Gamma(3/4)^2}{\pi}.$$

(c) For $0 < t < 1$ and $w > 0$, $p_t(x, -y; 0, -w)/p_t(x, -y; 0, w) = \exp(4w(3x - yt)/t^2)$ (the ratio of Lemma 9.1 and of (78)), which exceeds 1 when $x > 0$ and $y \leq 3x$; all weights are positive. For $y = 0$ the integrand $W(s, w) (G_a(-w) - G_a(w))$ of $\int_{x>a} \bar{h}_\star(x, 0) dx$ is therefore positive.

Numerical checks (write, 5 October 2026; uncertified floating point): an independent scipy quadrature of $\bar{h}_\star$ gives mass 1.000000000000 and $f_\star(0) = 0.390276218869$, both to 12 digits; mpmath confirms the Euler-integral step, the two quadratic values and ${}_3F_2 = 32/27$ to 40 digits. $\square$

Remark 4.4 ([write], Lemma 4.1 for the true density). Let $F_a^*(y) = \int_{x>a} \bar{h}_*(x, y) dx$. The assertions of Lemma 4.1 hold for F_a^* and $f_* = F_0^*$: joint continuity in $(a, y) \in [0, \infty) \times \mathbb{R}$, a common bound, and $0 < f_*(0) < \infty$.

Proof. The delivered proof does not transfer as it stands. Its majorant $C s^{3/4}(1-s)^{-1/2}$ becomes $C s^{-5/4}(1-s)^{-1/2}$ with $s^{-5/2}$, which is not integrable at $s = 0$. (The delivered check “dominating s power $3/4$ ” verifies the printed display only.) By the proof of Proposition 4.3, $F_a^*(y) = K_* \int_0^1 \int_0^\infty W(s, w) (G_a(-w) - G_a(w)) dw ds$ for every a and y . Here $G_a(\pm w)$ is continuous in (a, y) by its closed form, and the integrand is dominated by $K_* M(s, w)$, which does not depend on (a, y) . Dominated convergence gives joint continuity and the bound $K_* \int \int M$. Positivity is Proposition 4.3(c), and finiteness is (b).

Theorem 6.1 uses the density of μ only through Lemma 4.1 and through the fact that $F_a(y)$ is the density of $\mu(\{x > a\} \times dy)$. With $\bar{h}_*$ in place of $\bar{h}$, both hold, so Theorem 6.1, Section 7 and Theorem 1.1 hold with $f_Y = f_*$. The positivity on rectangles used in Lemma 9.1 ($2 < x < 3$, $1 < y < 2$) and in Part II, Section 16 (area large and velocity near 0) lies in the region $y \leq 3x$ of Proposition 4.3(c). It holds for $\bar{h}_*$ with the delivered arguments unchanged.

Remark 4.5 ([write], what the misprint affects). *Not affected:* the existence of one amplitude $A > 0$ for both parities and $D_n/C_n \rightarrow e^{-\lambda}$ (Sections 7–8 use only the existence of the limit in (36)); the enclosures of λ and r ; the harmonic identity (51), in which $f_Y(0)$ cancels between (50) and $B = rA$; Proposition 10.1 and Corollary 10.2, which use only (2); every upper bound of both Parts (survival (16) and atom bounds only); Part II’s Theorem 13.1 upper bounds, inverse and Proposition 18.1. *Wrong:* the value (22) as the marginal of μ , and the numerical reading of (6) with it (Remark 4.2). *Conditional on the density’s shape, now settled:* Lemma 4.1 (and through it Theorem 6.1 and the equivalents (2)), the rectangle of Lemma 9.1, and Part II’s lower bound in Section 16. They hold for $\bar{h}_*$ (Remark 4.4).

Why $\bar{h}_$ is the density of μ .* It is the function that [1, Remark 2] names, and the proof of [1, (8)] rests on [1, Lemma 16], the convergence $\bar{p}_1(x, y; u, v)/h(x, y) \rightarrow \kappa \bar{h}(u, v)$ of the killed transition density (dated note, 5 October 2026: the first wording, “ $\rightarrow \bar{h}(u, v)$ ”, omitted the factor κ of [1, Lemma 16, p. 186]), which is the same identification; Bär, Duraj and Wachtel [3, Propositions 3–4, Corollary 5] restate it with Groeneboom–Jongbloed–Wellner’s $\bar{h}$. Proposition 4.3(a) gives mass exactly 1, a non-trivial identity in $\Gamma(1/4)$ that the printed display fails. Numerically, a dynamic programme for the conditioned geometric-minus-one walk, made by the independent check of the write (floating point on a truncated state space, discarded mass below 10^{-9} , $n \leq 600$, extrapolated in powers of $n^{-1/2}$), gives $\sigma\sqrt{n} \mathbb{P}_{(1,1)}(V_n = 0 \mid \tau > n) \rightarrow 0.390277$, against $f_*(0) = 0.390276$ and 0.3267 for the printed display renormalized to mass one; ratios $1.1280, 0.3386, 0.0717$ of the conditional velocity density at $y = 1, 2, -1$ to its value at 0, against $1.1282, 0.3379, 0.0713$ for f_* and $1.8039, 0.1797, 0.0135$ for the renormalized printed display; and $\mathbb{E}[I_n/(\sigma n^{3/2}) \mid \tau > n] \rightarrow 0.57365$, against $\iint x \bar{h}_* = \frac{12}{5\sqrt{2}} \Gamma(3/4)/\Gamma(1/4) = 0.573587$ (proved below) and 0.217262 for the renormalized printed display. The same programme gives $n^{1/4} \mathbb{P}_{(1,1)}(\tau > n) \rightarrow c_{(1,1)} \approx 0.83360$, so $2^{1/4} c_{(1,1)} f_*(0) \approx 0.38689$, against $A \approx 0.3868892$ from the exact counts to $n = 500$. These extrapolations are evidence only. (Dated note, 5 October 2026: these sentences replace the write’s Monte Carlo evidence, two of whose values were wrong. The first wording read “Numerically, the velocity marginal of $\bar{h}_*$ has $f_*(1)/f_*(0) = 1.128$, $f_*(2)/f_*(0) = 0.338$ and $f_*(-1)/f_*(0) = 0.0713$. A Monte Carlo of the conditioned Gaussian walk at $n = 2000$ at the placement gave $1.107, 0.347$ and 0.0775 , and the printed display renormalized to mass one gives $1.803, 0.180$ and 0.0135 . The Monte Carlo conditional velocity density of the geometric walk near 0 was 0.448 at $n = 500$ and 0.395 at $n = 2000$, against $f_*(0) = 0.390$. Finally $2^{1/4} c_{(1,1)} f_*(0) = 0.388$ and 0.391 for the Monte Carlo values 0.8355 and

0.8426 of $c_{(1,1)}$, against $A \approx 0.387$ from the counts. The Monte Carlo values drift with n and are evidence only.” The value 0.448 at $n = 500$ was wrong: the dynamic programme gives 0.396060 at $n = 500$ (0.404539 at $n = 100$, decreasing in n). The value $c_{(1,1)} = 0.8426$ at $n = 2000$ lies about 0.009 above the dynamic programme’s values, which decrease to 0.83360 (0.835866 at $n = 600$), so the product 0.391 formed with it is too large; 0.8355 at $n = 500$ agrees with 0.836094.)

The mean area of $\bar{h}_\star$ (dated note, 5 October 2026; the identity is the independent check’s, the integrability argument the write’s). By (19), $p_t(x, y; u, v) = p_t(u, -v; x, -y)$: the substitution $(u - x - ty, v - y) \mapsto (x - u + tv, v - y)$ leaves the quadratic form unchanged. Hence $q_u(x, -y; 0, -w) = g_w(x, y) - g_{-w}(x, y)$, where $g_a(x, y) = p_u(0, a; x, y)$ is the density of $(X, V) = (au + \int_0^u B, a + B_u)$, a Gaussian vector with mean (au, a) , variances $u^3/3$ and u , and covariance $u^2/2$. Since (X, V) from $(0, -w)$ has the law of $-(X, V)$ from $(0, w)$,

$$\int_{\mathbb{R}} \int_{x>0} x q_u(x, -y; 0, -w) dx dy = \mathbb{E}_w[X; X > 0] + \mathbb{E}_w[X; X < 0] = \mathbb{E}_w X = wu.$$

For Fubini: $\iint |x| g_{\pm w} \leq wu + (u^3/3)^{1/2} \leq w + 1$ for $u \leq 1$; and since $q_u(x, -y; 0, -w) = \int_{-w}^w \partial_a g_a(x, y) da$, Cauchy–Schwarz with the Fisher information $4/u$ of the mean shift $a \mapsto (au, a)$ gives $\iint |x| \partial_a g_a \leq (a^2 u^2 + u^3/3)^{1/2} (4/u)^{1/2} \leq 2(|a| + 1)u^{-1/2}$. So $\iint_{x>0} x |q_u(x, -y; 0, -w)| dx dy \leq \min\{2(w+1), 4w(w+1)u^{-1/2}\}$, and with $u = 1 - s$ the weight $W(s, w)$ of the proof of Proposition 4.3 times this bound is integrable over $(0, 1) \times (0, \infty)$, exactly as M there (for $s \leq 1/2$ the w -integral is $O(s^{-3/4})$). With $\int_0^\infty w^{5/2} e^{-2w^2/s} dw = \frac{1}{2} \Gamma(7/4) (s/2)^{7/4}$,

$$\begin{aligned} \int_0^\infty \int_{\mathbb{R}} x \bar{h}_\star(x, y) dy dx &= K_\star \int_0^1 \int_0^\infty w^{5/2} s^{-5/2} (1-s) e^{-2w^2/s} dw ds \\ &= K_\star \frac{\Gamma(7/4)}{2^{11/4}} \int_0^1 s^{-3/4} (1-s) ds = K_\star \frac{\Gamma(7/4)}{2^{11/4}} \cdot \frac{16}{5} \\ &= \frac{12}{5\sqrt{2}} \frac{\Gamma(3/4)}{\Gamma(1/4)} = 0.5735865569 \dots, \end{aligned}$$

using $\Gamma(7/4) = \frac{3}{4} \Gamma(\frac{3}{4})$. The write checked the identity by direct two-dimensional quadrature of (19) at five pairs (u, w) (agreement with wu to 12 digits) and the last double integral to 12 digits (uncertified floating point).

Literature. A search on 5 October 2026 found no erratum to [1] and no published remark on this display. [3] prints $\bar{h}_{\text{GJW}}$ with the factor $p_s(0, w; 0, 0)$ and characterizes the meander density through it, consistent with $\bar{h}_\star$, but does not comment on the display in [1, Theorem 1]. The write reads [2] in the authors’ preprint of 30 August 1999, whose equation numbers agree with the citation in [1]. The misprinted display is that of the journal version only: arXiv:1207.2270v1 (10 July 2012), the only arXiv version of [1], states in its Theorem 1 a different density, $Ch(x, y)g_1(0, 0; x, y)$ on $\mathbb{R}_+ \times \mathbb{R}$ in that version’s notation, with no explicit integral. Whether that form is correct was not tested. (Dated note, 5 October 2026, from the independent check of the write.)

5 Free endpoint masses and a maximal estimate

Put $T_j = X_1 + \dots + X_j$. The ordinary span-one lattice local central limit theorem gives, uniformly over integers d ,

$$\sqrt{\ell} \mathbb{P}(T_\ell = d) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{d^2}{2\sigma^2\ell}\right) + o(1). \quad (24)$$

In particular $\sup_d \mathbb{P}(T_\ell = d) \leq C\ell^{-1/2}$ for $\ell \geq 1$. The hypotheses here are just the finite variance and span one already checked. For this particular increment law the exact atom is also explicit:

$$\mathbb{P}(T_\ell = d) = \binom{2\ell + d - 1}{\ell - 1} 2^{-2\ell - d} \quad (d \geq -\ell), \quad (25)$$

and it is zero otherwise. The ratio of consecutive positive masses is $(2\ell + d)/(2(\ell + d + 1))$, so $d = -1$ is a mode and the maximum is $\frac{1}{2} \binom{2\ell - 2}{\ell - 1} 4^{-(\ell - 1)} = O(\ell^{-1/2})$. Thus the uniform atom bound can alternatively be checked directly by Stirling.

Lemma 5.1. *There is a constant C , depending only on the increment law, such that for $\ell \geq 4$, $H > 0$, and an integer d with $|d| \leq H/2$,*

$$\mathbb{P}\left(T_\ell = d, \max_{j \leq \ell} |T_j| \geq H\right) \leq C \frac{\sqrt{\ell}}{H^2}. \quad (26)$$

Proof. Split at $h = \lfloor \ell/2 \rfloor$. For a violation in the first half, condition on the first h increments and bound the remaining independent endpoint atom by $C/\sqrt{\ell - h}$. Kolmogorov's inequality bounds the first-half maximum probability by $\sigma^2 h/H^2$, giving $C\sqrt{\ell}/H^2$.

For a violation at a time $j > h$, the equality $T_\ell = d$ implies $|T_\ell - T_j| \geq H - |d| \geq H/2$. The reversed last $\ell - h$ increments therefore have a partial sum of absolute value at least $H/2$. Condition on that suffix, use the first-half endpoint-atom bound $C/\sqrt{h}$, and then Kolmogorov's inequality for its reversed partial sums. This contributes another $C\sqrt{\ell}/H^2$. Their sum proves the estimate. $\square$

The estimate concerns joint endpoint mass. Dividing by an endpoint probability would introduce an unnecessary, possibly nonuniform bridge constant.

[write] **5 October 2026.** Part II, Lemma 15.2 (bounded bridge), proves the complementary lower bound by the same Kolmogorov split; Lemma 5.1 is the endpoint-weighted upper form. The atom formula (25) and its mode are Part II's (69) and Lemma 15.1, which adds a local lower bound.

6 Smoothing to one fixed terminal velocity

Theorem 6.1. *For every fixed positive-coordinate integer state $z = (a, v)$ and every fixed integer b ,*

$$\mathbb{P}_z(\tau > n, V_n = b) \sim \frac{c_z f_Y(0)}{\sigma} n^{-3/4}. \quad (27)$$

Also, uniformly over integer b ,

$$\mathbb{P}_z(\tau > n, V_n = b) \leq C_z (n+1)^{-3/4} \quad (n \geq 1). \quad (28)$$

Proof. For (28), split at $\lfloor n/2 \rfloor$, retain survival only in the first block, and use the free endpoint-atom bound in the second. The survival equivalent (16), enlarged to a global upper bound, then gives $C_z n^{-1/4} n^{-1/2}$. Enlarge C_z for small times.

For the equivalent fix $0 < \varepsilon < 1/2$ and put

$$k = \lfloor (1 - \varepsilon)n \rfloor, \quad \ell = n - k, \quad \alpha = 1 - \varepsilon.$$

All limits in n below keep ε fixed.

Upper bound. Dropping the area tests after k gives

$$\mathbb{P}_z(\tau > n, V_n = b) \leq \mathbb{E}_z[\mathbf{1}_{\{\tau > k\}} \mathbb{P}(T_\ell = b - V_k)]. \quad (29)$$

The uniform error in (24) is $o(\ell^{-1/2})$; after averaging it is multiplied by $\mathbb{P}_z(\tau > k) = O(k^{-1/4})$. Under the scaling $V_k = \sigma\sqrt{k}y$, the Gaussian test functions converge uniformly on $\mathbb{R}$ to the Gaussian with exponent $-\alpha y^2/(2\varepsilon)$. Thus (16)–(17) show that $n^{3/4}$ times the right side of (29) converges to

$$U_\varepsilon = \frac{c_z \alpha^{-1/4}}{\sigma \sqrt{2\pi\varepsilon}} \int_{\mathbb{R}} e^{-\alpha y^2/(2\varepsilon)} f_Y(y) dy. \quad (30)$$

Writing $t = \varepsilon/\alpha$ and $\phi_t(y) = (2\pi t)^{-1/2} e^{-y^2/(2t)}$, this is

$$U_\varepsilon = \frac{c_z}{\sigma} \alpha^{-3/4} \int_{\mathbb{R}} \phi_t(y) f_Y(y) dy \longrightarrow \frac{c_z f_Y(0)}{\sigma} \quad (\varepsilon \downarrow 0)$$

by Lemma 4.1. This proves the limsup bound.

Lower bound before the killing error. Fix $\delta, K > 0$ and at time k retain only the states

$$I_k > \delta n^{3/2}, \quad |V_k| < K\sqrt{n}. \quad (31)$$

First ignore killing in the last block. The same calculation gives the limit

$$L_{\varepsilon, \delta, K} = \frac{c_z \alpha^{-1/4}}{\sigma \sqrt{2\pi\varepsilon}} \int_{|y| < K/(\sigma\sqrt{\alpha})} e^{-\alpha y^2/(2\varepsilon)} F_{\delta/(\sigma\alpha^{3/2})}(y) dy. \quad (32)$$

Indeed, in conditional scaled coordinates the retained set has area threshold $\delta/(\sigma(k/n)^{3/2})$ and velocity cutoff $K/(\sigma\sqrt{k/n})$. The three limiting boundary lines are μ -null because μ has a density. Weak convergence therefore applies to the bounded Gaussian times this indicator. Moving boundary strips caused by floors have vanishing mass by Portmanteau. No uniform weak-limit rate in the parameters is being used.

Endpoint mass of paths killed in the last block. For a retained state (a, v) , the area after j more steps is

$$a + jv + \sum_{i=1}^j T_i.$$

For large n , $\ell \leq 2\varepsilon n$. If $\varepsilon \leq \delta/(8(K+1))$, then a failure of positivity implies

$$\max_{i \leq \ell} |T_i| \geq H, \quad H = \frac{\delta}{4\varepsilon} \sqrt{n}. \quad (33)$$

In fact $a - \ell|v| > \delta n^{3/2} - 2\varepsilon K n^{3/2} \geq \delta n^{3/2}/2$, so a nonpositive area requires $\max |T_i| \geq \delta n^{3/2}/(2\ell) \geq H$. The endpoint condition is $T_\ell = b - v$ and eventually $|b - v| \leq (K+1)\sqrt{n} \leq H/2$. Lemma 5.1 therefore gives, uniformly over all retained states,

$$\mathbb{P}_{(a,v)}(\tau \leq \ell, V_\ell = b) \leq C\varepsilon^{5/2} \delta^{-2} n^{-1/2}. \quad (34)$$

The constant here does not grow as $\varepsilon \downarrow 0$; the threshold from which n is sufficiently large may depend on $b, \delta, K, \varepsilon$. Multiplying by $\mathbb{P}_z(\tau > k)$ shows that the error in $n^{3/4}$ -scaled mass is at most $C_z \alpha^{-1/4} \varepsilon^{5/2} \delta^{-2}$ in the limit. Consequently

$$\liminf_{n \rightarrow \infty} n^{3/4} \mathbb{P}_z(\tau > n, V_n = b) \geq L_{\varepsilon, \delta, K} - C_z \alpha^{-1/4} \varepsilon^{5/2} \delta^{-2}. \quad (35)$$

Taking the limits in order. In (32) substitute $y = \sqrt{\varepsilon/\alpha} u$ to obtain

$$L_{\varepsilon,\delta,K} = \frac{c_z}{\sigma} \alpha^{-3/4} \int_{|u| < K/(\sigma\sqrt{\varepsilon})} \frac{e^{-u^2/2}}{\sqrt{2\pi}} F_{\delta/(\sigma\alpha^{3/2})}(\sqrt{\varepsilon/\alpha} u) du.$$

For fixed $\delta, K > 0$, joint continuity and the common bound of Lemma 4.1 give $L_{\varepsilon,\delta,K} \rightarrow (c_z/\sigma) F_{\delta/\sigma}(0)$ as $\varepsilon \downarrow 0$. The error in (35) vanishes. Finally let $\delta \downarrow 0$, giving $F_{\delta/\sigma}(0) \rightarrow f_Y(0)$. This agrees with the upper bound and proves (27). The order is $n \rightarrow \infty$, then $\varepsilon \downarrow 0$ at fixed δ, K , then $\delta \downarrow 0$. Any one fixed $K > 0$ suffices. $\square$

[write] **5 October 2026.** Theorem 6.1 contains Part II's Lemma 16.1 (order $k^{-3/4}$ from $(1,1)$). Part II's proof (a high-area rectangle at halfway time and a bounded bridge) is a second route to the order of magnitude. Here f_Y is to be read as $f_\star$, the velocity marginal of the true density; the proof uses only Lemma 4.1, which holds for it (Remark 4.4).

7 The common amplitude for all self-complementary counts

Apply Theorem 6.1 to each fixed h in (13). For both terminal values $r_n = -1, 0$,

$$m^{3/4} \mathbb{P}_{(1,1)}(\tau > m, V_m = r_n + 1 - h) \rightarrow \frac{c_{(1,1)} f_Y(0)}{\sigma}.$$

The uniform bound (28) gives a summable majorant after multiplication by 2^{-h-1} . Dominated convergence and $\sum_{h \geq 0} 2^{-h-1} = 1$ give

$$2^{-n} C_n \sim \frac{c_{(1,1)} f_Y(0)}{\sigma} m^{-3/4}. \quad (36)$$

Since $m \sim n/2$ along either parity, its amplitude is

$$A = \frac{2^{3/4} c_{(1,1)} f_Y(0)}{\sigma} = 2^{1/4} \kappa V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}) f_Y(0).$$

The half-length conversion contributes $2^{3/4}$, the span-one endpoint mass contributes $1/\sigma$, and the final-slack weights contribute total mass one. There is no residual parity multiplier. This proves the first assertion of Theorem 1.1.

[write] **5 October 2026.** With $f_\star(0)$ from Proposition 4.3, $A = 2^{1/4} c_{(1,1)} f_\star(0) = 0.46411 \dots \times c_{(1,1)}$. No certified value of $c_{(1,1)} = \kappa V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$ is known (Section 12.1); the counts suggest $c_{(1,1)} \approx 0.834$.

8 The strong equivalent from an exact convolution

Let S_k and T_k count ordinary all and strong tournament score sequences, with $S_0 = 1, T_0 = 0$. Write their ordinary generating functions as $S(z), T(z)$, and use $C(z), D(z)$ for the self-complementary counts. Stockmeyer's Theorems 11 and 13 [5] give

$$S(z) = \frac{1}{1 - T(z)}, \quad (37)$$

$$D_n = C_n - \sum_{1 \leq k \leq \lfloor n/2 \rfloor} T_k C_{n-2k} \quad (n \geq 1), \quad (38)$$

$$D(z) = C(z)(1 - T(z^2)), \quad C(z) = D(z)S(z^2). \quad (39)$$

These are formal power-series identities, so no analytic assumption is needed at this stage. The established ordinary-score asymptotic in [6] in particular yields

$$0 \leq T_k/4^k \leq S_k/4^k \leq C(k+1)^{-5/2}. \quad (40)$$

This ordinary-score input is not a new result of the present report.

[write] **5 October 2026.** S_k and T_k are the S_n and I_n of **a000571-tournament-score-sequences** (strong ordinary score sequences there have $I_0 = 0$, like $T_0 = 0$ here), and $1 - T(z) = e^{-A(z)}$ in its notation. Why the strong fraction is $e^{-\lambda}$ here but $I_n/S_n \rightarrow e^{-2\lambda}$ there is explained in the front matter (“Relation to the ordinary score sequences”).

8.1 Justifying the convolution limit

Put $c_n = C_n/2^n$ and $t_k = T_k/4^k$. We have $c_n \sim An^{-3/4}$ and hence $c_n \leq B(n+1)^{-3/4}$ for all $n \geq 0$. For fixed K ,

$$n^{3/4} \sum_{k=1}^K t_k c_{n-2k} \longrightarrow A \sum_{k=1}^K t_k.$$

For $K < k \leq n/4$, the bound on c_{n-2k} gives

$$n^{3/4} \sum_{K < k \leq n/4} t_k c_{n-2k} \leq C \sum_{k > K} t_k \longrightarrow 0 \quad (K \rightarrow \infty), \quad (41)$$

uniformly in large n . The remaining terms satisfy

$$\begin{aligned} n^{3/4} \sum_{n/4 < k \leq n/2} t_k c_{n-2k} &\leq C n^{3/4} n^{-5/2} \sum_{0 \leq j \leq n/2} (j+1)^{-3/4} \\ &= O(n^{-3/2}). \end{aligned} \quad (42)$$

Only one parity of $j = n - 2k$ occurs, and extending the sum only enlarges it. Thus

$$n^{3/4} \sum_{1 \leq k \leq n/2} t_k c_{n-2k} \longrightarrow A \sum_{k \geq 1} t_k. \quad (43)$$

Using (38),

$$D_n \sim A(1 - T(1/4)) 2^n n^{-3/4} = \frac{A}{S(1/4)} 2^n n^{-3/4}. \quad (44)$$

The ordinary tail bound makes $S(1/4)$ finite. Taking increasing real limits in (37) gives $1 - T(1/4) = 1/S(1/4) > 0$.

8.2 Identifying the ratio and its numerical enclosure

The ordinary recurrence and its generating-function form, recorded in [6, 7], are

$$kS_k = \sum_{j=1}^k N_j S_{k-j}, \quad S(z) = \exp \left(\sum_{k \geq 1} \frac{N_k}{k} z^k \right), \quad (45)$$

with the exact divisor formula (4). Its terms at $z = 1/4$ are $O(k^{-5/2})$: the $d = k$ term has this order after division by $k4^k$, and all proper divisors satisfy $d \leq k/2$, giving an exponentially smaller bound. Thus the increasing real limit gives $S(1/4) = e^\lambda$ and proves (2) with $r = e^{-\lambda}$.

Kolesnik [6, Section 3.1, equation (8)] obtains the certified interval

$$0.330237542 \leq \lambda \leq 0.330237546. \quad (46)$$

Exponentiating with the reversed order yields (5). The archived checker also independently reproduces a narrower enclosure by computing $N_1, \dots, N_{1669}$ exactly and combining their rational partial sum with the tail estimate of [6, Lemma 11, equation (17)]. Writing $K = 1669$, that imported estimate is

$$\frac{1 - 2/K}{3\sqrt{\pi}K^{3/2}} \leq \sum_{k>K} \frac{N_k}{k4^k} \leq \frac{1}{3\sqrt{\pi}K^{3/2}}.$$

Rational bounds for π , square roots and the exponential, with outward rounding, certify in particular

$$\begin{aligned} 0.3302375420437401 &\leq \lambda \leq 0.3302375453489038, \\ 0.7187529767248666 &\leq r \leq 0.7187529791004629. \end{aligned}$$

The exact partial sum and full outward enclosures appear in `checks/results.json`. This certificate uses a published infinite-tail theorem; finite enumeration alone would not certify λ . We have $1/2 < r < 1$ and $D_n/C_n \rightarrow r$.

[write] **5 October 2026.** The value $\lambda = 0.3302375439859293$ of `a000571-tournament-score-sequences` lies inside the enclosure above, and $e^{-\lambda} = 0.7187529777045\dots$ at that value. Floating-point counts at the placement give $D_n/C_n = 0.6964, 0.7082, 0.7137, 0.7154, 0.7163$ at $n = 40, 80, 160, 240, 320$, approaching r with a gap of order $1/n$. This is evidence, not a proof; the identity $D(z) = C(z)(1 - T(z^2))$ was checked exactly for $n \leq 44$.

These numerical enclosures concern λ and r . Formula (6) characterizes A exactly, but contains a discrete harmonic function that has not been evaluated or numerically enclosed here. The limiting density is explicit; that does not by itself numerically determine A .

[write] **5 October 2026.** The density as printed in Section 4 is not that of μ (Remark 4.2); the corrected factor $f_*(0) = \sqrt{2/3} \Gamma(3/4)^2/\pi$ is explicit (Proposition 4.3), and $A = \sqrt{6} \Gamma(3/4) \pi^{-3/2} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$. The discrete harmonic function remains unevaluated, as stated.

9 A direct strong amplitude and a harmonic identity

This section gives an independent expression for the strong amplitude. It is optional for Theorem 1.1, whose proof is already complete. It also checks that a mixture over unbounded starting states can be justified without assuming a uniform version of (16).

Lemma 9.1. *There is a constant C such that, for all integers $g, m \geq 1$,*

$$\mathbb{P}_{(g,g)}(\tau > m) \leq C\sqrt{g}m^{-1/4}, \quad (47)$$

$$\sup_{b \in \mathbb{Z}} \mathbb{P}_{(g,g)}(\tau > m, V_m = b) \leq C\sqrt{g}m^{-3/4}. \quad (48)$$

In particular $c_{(g,g)} \leq C\sqrt{g}$.

Proof. From $(1, 1)$, use time $N = g^2$ and the normalized rectangle

$$2 < I_N/(\sigma N^{3/2}) < 3, \quad 1 < V_N/(\sigma\sqrt{N}) < 2.$$

It has positive μ -mass. Indeed,

$$\frac{p_t(x, -y; 0, -w)}{p_t(x, -y; 0, w)} = \exp\left(\frac{4w(3x - yt)}{t^2}\right) > 1$$

for $0 < t < 1$ on this rectangle, so (18) is positive there. By (16)–(17), the probability of reaching it while surviving is at least $cg^{-1/2}$ for all sufficiently large g . Every state there dominates (g, g) coordinatewise. For the finitely many smaller g , force $G_1 = g$ and $G_j = 1$ for $2 \leq j \leq g^2$. Starting from $(1, 1)$ the velocity then equals g from time one onward and the area stays positive; its time- g^2 state dominates (g, g) . Each forced path has positive probability, so decreasing c covers all g .

The common-increment coupling is monotone in both starting coordinates. The Markov property therefore gives

$$\mathbb{P}_{(1,1)}(\tau > g^2 + m) \geq cg^{-1/2}\mathbb{P}_{(g,g)}(\tau > m).$$

The global upper bound following from (16) proves (47). Split an endpoint event at halfway and use the free atom bound to obtain (48), enlarging C for $m = 1$. Taking $m \rightarrow \infty$ in the first bound gives the assertion about $c_{(g,g)}$. $\square$

[write] **5 October 2026.** The positivity on the rectangle uses only the ratio displayed in the proof and positive weights, so it holds for the true density $\bar{h}_*$ (Proposition 4.3(c)); the rectangle lies in $y \leq 3x$.

For strict Landau inequalities the first geometric value must be $g \geq 1$. At time one its unshifted state is $(A_1, V_1) = (g, g)$. Decomposing this first value and the final slack yields, for $m \geq 2$,

$$2^{-n}D_n = \sum_{g \geq 1} \sum_{h \geq 0} 2^{-g-1} 2^{-h-1} \mathbb{P}_{(g,g)}(\tau > m-1, V_{m-1} = r_n + 1 - h). \quad (49)$$

Here the probabilities use the state after that first step as their new start. Theorem 6.1 applies for each fixed g, h ; Lemma 9.1 and $\sum_{g \geq 1} 2^{-g-1} \sqrt{g} < \infty$ justify summation. Hence the direct strong amplitude is

$$B = \frac{2^{3/4} f_Y(0)}{\sigma} \sum_{g \geq 1} 2^{-g-1} c_{(g,g)}. \quad (50)$$

Comparing with $B = rA$ gives the exact identity

$$\sum_{g \geq 1} 2^{-g-1} V_{\text{DW}}(g/\sqrt{2}, g/\sqrt{2}) = e^{-\lambda} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}). \quad (51)$$

The first-step weights are unconditional; their sum over $g \geq 1$ is $1/2$. Renormalizing them would incorrectly double the amplitude. The identity is a consequence of two independently justified amplitude representations; whether it has a shorter direct harmonic or combinatorial proof is a further question.

[write] **5 October 2026.** $f_Y(0)$ cancels between (50) and $B = rA$, so (51) does not depend on its value and is unaffected by the misprint of Section 4.

10 Smooth inverses and integer thresholds

All logarithms in this section are natural. Put $a = 3/4$, $\gamma = \log 2$, and $A_C = A$, $A_D = rA$. For $E = C$ or D , define

$$F_E(t) = A_E e^{\gamma t} t^{-a} \quad (t > 0), \quad N_E(X) = \min\{n \geq 1 : E_n \geq X\}.$$

The counts tend to infinity, so the minimum exists. Their equivalents imply $E_{n+1}/E_n \rightarrow 2$, hence eventual strict increase; a finite initial exception such as $D_1 > D_2$ has no effect for large X .

[write] **5 October 2026.** Part II, Proposition 18.1, proves more for all n : C_n is nondecreasing for $n \geq 0$ and D_n for $n \geq 2$ (not strictly: $C_1 = C_2$ and $C_3 = C_4$).

Proposition 10.1. *For all sufficiently large X , the large real solution of $F_E(t) = X$ is*

$$q_E(X) = -\frac{a}{\gamma} W_{-1} \left(-\frac{\gamma}{a} \left(\frac{A_E}{X} \right)^{1/a} \right), \quad (52)$$

where W_{-1} is the real branch with values at most -1 . It satisfies

$$q_E(X) = \frac{\log X}{\gamma} + \frac{a}{\gamma} \log \log X - \frac{a}{\gamma} \log \gamma - \frac{\log A_E}{\gamma} + o(1). \quad (53)$$

Moreover $q_E(E_n) = n + o(1)$. There exists $\eta_E(X) \geq 0$ with $\eta_E(X) \rightarrow 0$ such that, for all sufficiently large X ,

$$[q_E(X) - \eta_E(X)] \leq N_E(X) \leq [q_E(X) + \eta_E(X)]. \quad (54)$$

In particular $N_E(X) = q_E(X) + O(1)$, but a global claim $N_E(X) = q_E(X) + o(1)$ is not justified and is false.

Proof. The function F_E is strictly increasing for $t > a/\gamma$. Rearranging its equation gives $te^{-\gamma t/a} = (A_E/X)^{1/a}$, from which (52) follows. The branch W_{-1} gives the large solution; W_0 would give a solution approaching zero.

Let $L = \log X$ and set

$$u = \frac{1}{\gamma} (L + a \log L - a \log \gamma - \log A_E).$$

Then $u \sim L/\gamma$ and $\log F_E(u) = \log A_E + \gamma u - a \log u = L + o(1)$. Since $(\log F_E)'(t) = \gamma - a/t$ is bounded away from zero at large t , the mean value theorem gives $q_E(X) - u = o(1)$, proving (53). Similarly $\log E_n - \log F_E(n) = o(1)$ proves $q_E(E_n) = n + o(1)$.

For the integer statement write $e_n = q_E(E_n) - n \rightarrow 0$ and, for large X , let $m = N_E(X)$. Eventual strict increase and removal of the finite initial range give $E_{m-1} < X \leq E_m$, hence

$$m - 1 + e_{m-1} < q_E(X) \leq m + e_m.$$

Take, for example, $\eta_E(X) = \max(|e_{m-1}|, |e_m|)$. It tends to zero because $m \rightarrow \infty$, and $q_E(X) - \eta_E(X) \leq m < q_E(X) + 1 + \eta_E(X)$. The two ceiling inequalities follow, including the case when a bound is an integer. These brackets are qualitative and not an effective computable error bound. To see why an $o(1)$ error for N_E cannot hold for all real X , put $X = F_E(j+1/2)$. Then $q_E(X) = j+1/2$, while its distance from any integer is $1/2$. For large j , the brackets in fact force $N_E(X) = j+1$. $\square$

In base-two form the smooth expansion is

$$q_E(X) = \log_2 X + \frac{3}{4} \log_2 \log_2 X - \log_2 A_E + o(1).$$

In particular

$$q_D(X) - q_C(X) \rightarrow \frac{\lambda}{\log 2} \in (0, 1). \quad (55)$$

The nonintegral limiting smooth displacement must not be confused with the integer threshold gap.

Corollary 10.2. *For every sufficiently large real X ,*

$$N_C(X) \leq N_D(X) \leq N_C(X) + 1. \quad (56)$$

Both possible gaps, zero and one, occur for infinitely many X tending to infinity, even if thresholds are restricted to integers.

Proof. Always $D_n \leq C_n$, and (2) gives $D_{n+1}/C_n \rightarrow 2r > 1$. For $m = N_C(X)$ large, therefore $D_{m+1} > C_m \geq X$, so $N_D(X) \leq m + 1$; the other inequality is immediate. At $X = D_n$ the eventual inequalities $C_{n-1} < D_n \leq C_n$ and strict growth of both counts give $N_C(X) = N_D(X) = n$. At $X = C_n$, we have eventually $D_n < C_n < D_{n+1}$, giving $N_C(X) = n$ and $N_D(X) = n + 1$. Both sequences of thresholds are integer and tend to infinity. $\square$

Neither a rate for (2) nor a higher-order coefficient expansion is used in these arguments. Further terms of the smooth Lambert inverse alone would not supply corresponding further terms for the actual count thresholds.

[write] **5 October 2026.** Part II's inverse (59), $N_E(x) = \log_2 x + \frac{3}{4} \log_2 \log_2 x + O(1)$, follows from (53) and (54) in base two; Proposition 10.1 identifies the constant $-\log_2 A_E$ of the smooth inverse. Numerically $\lambda/\log 2 = 0.47643206702\dots$ in (55). The same Lambert- W_{-1} mechanism, for $4^x x^{-5/2}$, is `tss:eq:lambert` of `a000571-tournament-score-sequences`.

11 Exact checks and reproducibility

The accompanying reader archive separates finite verification from the analytic argument. Exact finite counts can check an encoding, coefficient identity or decimal transformation; they cannot prove the survival theorem, its limiting density, or the leading equivalents.

The package contains the editable TeX, rendered PDF, frozen reference fixtures, executable arithmetic checks, mutation tests, a strict hash manifest, and scripts for replay and deterministic rebuilding. Source-paper PDFs and working research notes are not included. Primary references are linked below. The finite-check coverage and ranges are specified in `checks/reader_validation.md`, and the actual results are preserved in machine-readable files.

The reader workflow is:

1. Run `python3 replay.py`. It checks the exact declared file set, then runs the arithmetic, semantic/schema mutation and manifest-attack suites both normally and under `python3 -O`, comparing every result with the archived output.
2. Run `python3 replay.py -build-pdf` to additionally rebuild the PDF twice in independent temporary directories and compare both builds with the archived PDF byte for byte. The recorded TeX toolchain is needed for exact byte identity; another version may render the same mathematics differently.
3. Run `python3 make_archive.py -output /tmp/report116.zip` from an unchanged extracted package to recreate its deterministic ZIP bytes.

The scripts use explicit checks that remain active with optimization. Replay does not require network access and does not modify the sealed package. The manifest rejects unexpected or missing files and directories, links, special files, duplicate or unsafe paths, malformed entries and hash

changes. It is an integrity record, not a signature or a mathematical certificate. Theorem 1.1 rests on the proof and its stated published inputs.

[write] **5 October 2026.** The package is shipped under other names: `checks/reader_validation.md` is `116-leading-checks-reader_validation.md`, `checks/results.json` is `data/116-leading-checks-r`
`esults.json`, and the scripts are in `code/` (README, “Delivered path to shipped path”). The delivered checks of Section 4’s algebra, including “dominating s power 3/4” and the formula (22), verify the algebra of the printed display. They cannot detect the misprint, which is in the imported display itself (Remark 4.2).

12 Further questions and precise limits

1. **Evaluate and certify the amplitude.** Develop rigorous bounds for $V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$ in the specified normalization and combine them with controlled quadrature for $f_Y(0)$. The exact expression (6) and the harmonic identity (51) provide consistency checks. A fit to moderate counts would be evidence, not a certified value.
2. **Obtain quantitative errors and higher corrections.** The smoothing proof takes successive limits and imports weak convergence without a rate. A rate would require quantitative versions of the survival and conditional limit inputs, together with corresponding marginal and last-block estimates. Any correction analysis must track the two terminal values, the half-length conversion and the slack mixture separately before claiming that correction terms are parity-independent.
3. **Explain the harmonic identity directly.** Find a potential-theoretic or combinatorial proof of (51) that does not compare two asymptotic amplitude formulas. Such a proof could clarify its computational value or possible extensions to other increment laws and symmetry constraints.
4. **Make inverse bounds effective.** Certified bounds for the amplitude and counting error could turn (54) into usable threshold brackets. No explicit rounding cutoff is supplied here.

The established conclusions are the full leading equivalents, their common parity constants, the exact limiting strong fraction, the stated harmonic identity, and the qualitative inverse refinements. No numerical amplitude, convergence rate, higher-order counting coefficient or effective asymptotic threshold is asserted.

12.1 Further questions and research

[write] (5 October 2026.) Under Vladimir’s standing rule of 4 October 2026, this subsection collects every claim of Part I stated without proof, with its source, its sketch and what is missing, and the one claim refuted.

1. *Evaluate and certify the amplitude* (item 1 above). Re-scoped: $f_*(0)$ is now explicit (Proposition 4.3), so $A = \sqrt{6} \Gamma(3/4) \pi^{-3/2} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$ and the question is $V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$, equivalently $c_{(1,1)}$. Evidence only: $A \approx 0.387$ from the counts (Remark 4.2), hence $c_{(1,1)} \approx 0.834$ and $V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2}) \approx 0.718$; A dynamic programme for the walk gives $n^{1/4} \mathbb{P}_{(1,1)}(\tau > n) = 0.835866$ at $n = 600$, extrapolating to 0.83360 , and $2^{1/4} \cdot 0.83360 \cdot f_*(0) \approx 0.38689$, against $A \approx 0.3868892$ from parity fits to the counts to $n = 500$ (Remark 4.5). (Dated note, 5 October 2026: the first wording read “Monte Carlo gives $n^{1/4} \mathbb{P}_{(1,1)}(\tau > n) = 0.8355, 0.8426$ at $n = 500, 2000$.”; the value at $n = 2000$ lies about 0.009 above the dynamic programme’s

- decreasing values.) Missing: rigorous two-sided bounds on V_{DW} , for example from the corrector representation [1, Proposition 4], and a certified error for the counts.
2. *Rates and higher corrections* (item 2 above). The data show a parity-dependent correction of relative order about $n^{-1/2}$: $C_n n^{3/4} 2^{-n}$ increases on even n and decreases on odd n (Remark 4.2), and $D_n/C_n - r$ is of order $1/n$. This is an observation from the placement computations, not claimed by either Part. Missing: quantitative versions of (16)–(17) and of the smoothing step, tracking the terminal value, the half-length conversion and the slack mixture separately.
 3. *A direct proof of the harmonic identity* (51) (item 3 above). Open.
 4. *Effective inverse brackets* (item 4 above). Open; needs item 1 and an effective rate.
 5. *Correction expansions and transseries* (Part II, Section 19, third step). Open in both Parts.
 6. *The density of μ* . Settled to the standard of the cited literature (Remark 4.5): $\bar{h}_\star = \kappa^{-1} \bar{h}_{\text{GJW}}(1, x, -y)$. The write proves that $\bar{h}_\star$ has mass 1 and computes $f_\star(0)$; it does not re-derive [2, (2.24)] or [1, Lemma 16]. A self-contained check would derive the meander density at time 1 from the killed transition density of the Kolmogorov diffusion (Lachal’s formula, as in the proof of [1, Lemma 16]). (Dated note, 5 October 2026: the first wording of this sentence ended “, or compare $\bar{h}_\star$ with the conditioned walk more precisely than the Monte Carlo above.” That comparison is now made numerically: the independent check’s dynamic programme for the conditioned walk to $n = 600$ matches $f_\star(0)$ to six digits, the mean area $\iint x \bar{h}_\star$ to within 10^{-4} and the velocity ratios at $y = 1, 2, -1$ to within $7 \cdot 10^{-4}$, and rejects the renormalized printed display (Remark 4.5). This is evidence, not a derivation; the self-contained derivation remains open.) The velocity marginal $f_\star(y)$ at $y \neq 0$ has no closed form here.

Refuted (Remark 4.2): the identification of (22), $0.0377237\dots$, with the velocity marginal of μ at 0, and with it the evaluation of (6); the display (18) as the density of μ . The statements stay printed as delivered. Nothing else in this Part failed a check.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the write, made by the intake after the write (1b792e07a), reread Section 4.2, that is Remark 4.2, Proposition 4.3 with its proof, Remarks 4.4 and 4.5, and the front matter’s “Relation to the ordinary score sequences”. It found every mathematical claim valid, with no counterexample and no gap in any proof chain. It found four defects in the write’s supporting text; each is corrected where it stands, with a dated note keeping the first wording. The prediction with the printed density is $A \approx 0.0374$, not 0.038 (Remark 4.2). Two Monte Carlo values of Remark 4.5 were wrong: the conditional velocity density at 0 at $n = 500$ is 0.396060, not 0.448, and $c_{(1,1)} = 0.8426$ at $n = 2000$ is about 0.009 too large; all the Monte Carlo evidence is replaced by the check’s walk computation, and items 1 and 6 above are updated. The paraphrase of [1, Lemma 16] omitted the factor κ . The literature paragraph did not say that the misprint is specific to the journal version. The check also noted that the integral defining $\bar{h}_\star$ converges absolutely for every $x > 0$, not only almost everywhere, since $|q| \leq Cw$ near $s = 0$; the statement is kept, true and weaker. No other statement or number changed.

The check did not use the write’s scripts. It downloaded the published version of [1] from numdam and read pp. 169–170 and 186 as rendered pages: the display of Theorem 1 is exactly (18), Remark 2 identifies the density as $\kappa^{-1} \bar{h}_{\text{GJW}}(1, x, -y)$, and Lemma 16 reads $\kappa \bar{h}(u, v)$. It read [2, (2.24)] in the authors’ preprint of 30 August 1999 (p. 10 rendered) and confirmed that with $p_s(0, w; 0, 0) = \sqrt{3} e^{-2w^2/s}/(\pi s^2)$ it gives $\bar{h}_\star$, with $\kappa^{-1} \cdot 12/(\pi\sqrt{2\pi}) = K_\star$ to 30 digits; [3] is consistent. Its own quadrature of the printed display, after the closed-form w -integration, gives

mass 0.11545529600527263 (17 digits, agreeing with Remark 4.2) and $f(0) = 0.03772372389\dots$, equal to the delivered ${}_3F_2$ value (22) to 30 digits. It re-derived every step of Proposition 4.3(a)–(c) by hand; its own quadrature (after $s = r^4$) gives mass 1 and $f_\star(0) = \sqrt{2/3}\Gamma(3/4)^2/\pi$ to 28 digits, and ${}_3F_2 = 32/27$ to 30 digits. It recomputed the two bounds on q and the majorant M used in Remark 4.4. For the counts, it checked the OEIS b-files of A345470 ($n \leq 500$) and A351869 ($n \leq 501$) by brute force with the full Landau criterion for $n \leq 12$ and by its own exact big-integer half-sequence dynamic programme (every $n \leq 40$ and every seventh n to 160), with no mismatch, and reproduced the values $C_n n^{3/4} 2^{-n}$ of Remark 4.2. Per-parity least-squares fits on $n \in [150, 500]$, $[250, 500]$ and $[350, 500]$ with two to five correction terms in powers of $n^{-1/2}$, and a joint fit, all give $A = 0.3868892(2)$; the even and odd fits agree to 10^{-7} , with first corrections $\mp 0.2024 n^{-1/2}$. The decisive check was a dynamic programme of the conditioned geometric-minus-one walk from $(1, 1)$ to $n = 600$ (floating point on a truncated state space, discarded mass below 10^{-9}), extrapolated in powers of $n^{-1/2}$: it reproduces $f_\star(0)$, the shape of $f_\star$ and the mean area of $h_\star$, rejects the printed display, and closes the amplitude chain, $2^{1/4} c_{(1,1)} f_\star(0) \approx 0.38689$ against $A = 0.3868892(2)$ (Remark 4.5). For the mean area it derived $\iint_{x>0} x q_u = wu$ (and $\iint_{x>0} y q_u = w$); the write added the integrability argument and checked the identity by quadrature. It confirmed the front matter’s explanation of the strong fraction: $I(z) = 1 - e^{-A(z)}$, so $1 - I(1/4) = 1/S(1/4) = e^{-\lambda}$, and $D_{500}/C_{500} = 0.7172086$ approaches $e^{-\lambda} = 0.71875297770\dots$ with a gap of order $1/n$. This was a careful reading with independent computation, not a formal verification.

Part II

The growth theorem and monotonicity

Bundle Report 114, archive `Self_Complementary_Tournament_Scores_Growth_and_Inverse_Source.zip`, delivered as `report114.tex` (11 pages), dated 2 October 2026. Delivered title “The growth of self complementary tournament score sequences”; author line “Report 114”. Printed as delivered, with `[write]` notes and `\binom` for `\choose`. Delivered Section k is Section $k + 12$ here, its statement $k.j$ is $(k + 12).j$, and its equation (k) is $(k + 56)$. Section 21 is added in the write.

Abstract of this Part (as delivered). Let $C(n)$ count nondecreasing tournament score sequences invariant under score complementation, and let $D(n)$ count the strong sequences among them. These are OEIS A345470 and A351869. We prove

$$C(n) = \Theta\left(2^n n^{-3/4}\right), \quad D(n) = \Theta\left(2^n n^{-3/4}\right).$$

This establishes the two order-of-growth conjectures stated in Section 6 of Stockmeyer’s 2023 paper. The proof gives an exact encoding by integrated geometric random walks and combines the survival and conditional weak-limit theorem of Denisov and Wachtel with an elementary bounded-bridge estimate. It treats both parities, strict versus weak Landau inequalities, and the final untested slack step explicitly. The corresponding threshold inverses satisfy $\log_2 x + \frac{3}{4} \log_2 \log_2 x + O(1)$. Leading constants, an asymptotic strong-to-all ratio, and correction expansions are not established. Exact finite checks and a reproducible build accompany the proof, but are logically separate from it.

`[write]` **5 October 2026.** Part I (Report 116, written after this Part and crediting it for the Θ estimates) proves the leading equivalents with one amplitude for both parities, the strong-to-all ratio $e^{-\lambda}$ and an inverse with constant term. The sentence “Leading constants, an asymptotic strong-to-all ratio, and correction expansions are not established” remains true of this Part. The first two are established in Part I; correction expansions remain open in both Parts. The density displayed in Section 16 is the misprinted one of [1]; the argument here survives unchanged (Remark 4.4).

13 The problem and the result

The score of a tournament vertex is its outdegree. A score sequence is the nondecreasing list of these scores. Landau’s criterion says that an integer sequence $0 \leq s_1 \leq \dots \leq s_n$ is a tournament score sequence precisely when

$$\sum_{i=1}^r s_i \geq \binom{r}{2} \quad (1 \leq r < n), \quad \sum_{i=1}^n s_i = \binom{n}{2}. \quad (57)$$

A score sequence is *strong* precisely when all the proper inequalities in (57) are strict. These criteria are recorded in Stockmeyer [5, Propositions 1 and 7].

We call the score sequence self-complementary when

$$s_i + s_{n+1-i} = n - 1 \quad (1 \leq i \leq n). \quad (58)$$

Let $C(n)$ count score sequences satisfying (58), and let $D(n)$ count those that are also strong. With the empty sequence convention, $C(0) = D(0) = 1$; also $C(1) = D(1) = 1$.

The objects being counted. Each sorted score sequence is counted once, regardless of how many tournaments realize it. We do not count labeled tournaments or isomorphism classes of self-complementary tournaments. Complement symmetry of a score sequence does not imply that every tournament realizing it is isomorphic to its reverse. Stockmeyer discusses this distinction in Section 3 [5].

Theorem 13.1. *There are positive constants c_C, C_C, c_D, C_D and an integer n_0 such that, for every $n \geq n_0$,*

$$c_C 2^n n^{-3/4} \leq C(n) \leq C_C 2^n n^{-3/4}, \quad c_D 2^n n^{-3/4} \leq D(n) \leq C_D 2^n n^{-3/4}.$$

For $E = C$ or $E = D$, define

$$N_E(x) = \min\{n \geq 1 : E(n) \geq x\}.$$

Then, as $x \rightarrow \infty$,

$$N_E(x) = \log_2 x + \frac{3}{4} \log_2 \log_2 x + O(1). \quad (59)$$

Stockmeyer conjectured the two Θ estimates in [5, Section 6, p. 8]. The proof here uses a published probability theorem with its hypotheses checked below. This report makes no claim about publication, peer review, priority, or the absence of other proofs.

[write] **5 October 2026.** This is the first proof of the two Θ estimates. Part I, Theorem 1.1, implies them, and Proposition 10.1 implies (59) with $O(1)$ replaced by $-\log_2 A_E + o(1)$ for the smooth inverse (in base two, (53) reads $\log_2 X + \frac{3}{4} \log_2 \log_2 X - \log_2 A_E + o(1)$). The lower bounds here use the density of the limit law only through the positivity of Section 16, which holds for the true density (Remark 4.4); the upper bounds do not use the density at all.

13.1 Proof map

The exact encoding in Section 14 expresses each count as 2^n times an integrated-walk probability with a fixed terminal velocity. The probability has order $n^{-3/4}$: survival to a halfway time costs $n^{-1/4}$, and hitting one specified velocity over the remaining block costs $n^{-1/2}$. The lower bound must couple these two requirements without losing survival. A large positive halfway area and a bounded-velocity bridge accomplish that. Sections 15–16 supply the argument, and Section 17 transfers it back to both counting problems.

14 Exact encoding by a half sequence

Write $n = 2m$ or $n = 2m + 1$ and set

$$M = \left\lfloor \frac{n-1}{2} \right\rfloor = \begin{cases} m-1, & n = 2m, \\ m, & n = 2m+1. \end{cases}$$

For $m \geq 1$, a self-complementary candidate is uniquely determined by

$$0 \leq s_1 \leq \cdots \leq s_m \leq M. \quad (60)$$

Reflect these entries using (58); for odd n , insert the central score $s_{m+1} = m$. The bound M is exactly what makes the full reflected sequence nondecreasing. Its total is automatically $\binom{n}{2}$.

For the full sequence define the Landau excess

$$A_r = \sum_{i=1}^r s_i - \binom{r}{2}, \quad 0 \leq r \leq n.$$

The last r scores sum to $r(n-1) - \sum_{i=1}^r s_i$, so

$$\begin{aligned} A_{n-r} &= \binom{n}{2} - r(n-1) + \sum_{i=1}^r s_i - \binom{n-r}{2} \\ &= \sum_{i=1}^r s_i - \binom{r}{2} = A_r. \end{aligned}$$

Consequently all the Landau inequalities are equivalent to

$$A_r \geq 0 \quad (1 \leq r \leq m), \quad (61)$$

and strongness is equivalent to $A_r > 0$ on exactly the same index set. This includes the even midpoint and, by symmetry, both odd midpoint inequalities. There is no omitted middle condition.

14.1 Differences and the final slack

Introduce nonnegative integers

$$G_1 = s_1, \quad G_j = s_j - s_{j-1} \quad (2 \leq j \leq m), \quad G_{m+1} = M - s_m. \quad (62)$$

Then

$$G_1 + \cdots + G_{m+1} = M. \quad (63)$$

Conversely, each nonnegative composition (63) gives (60) by $s_r = G_1 + \cdots + G_r$. Thus this is a bijection, not an approximation or a weighting of score sequences by realizations.

Now take $G_1, \dots, G_{m+1}$ independently with

$$\mathbb{P}(G_j = g) = 2^{-g-1}, \quad g = 0, 1, 2, \dots \quad (64)$$

Every composition (63) has probability

$$2^{-(M+m+1)} = 2^{-n}$$

for either parity. Define an ordinary walk and its integrated area by

$$X_j = G_j - 1, \quad V_0 = 1, \quad V_r = 1 + \sum_{j=1}^r X_j, \quad A_0 = 0, \quad A_r = \sum_{j=1}^r V_j. \quad (65)$$

For $r \leq m$, $V_r = s_r - r + 1$, so this A_r is precisely the Landau excess. The total-composition condition becomes

$$V_{m+1} = 1 + M - (m+1) = M - m.$$

We have proved the following exact identities.

Proposition 14.1. *For every $m \geq 1$,*

$$C(2m) = 2^{2m} \mathbb{P}(A_r \geq 0 \ (1 \leq r \leq m), \ V_{m+1} = -1), \quad (66)$$

$$C(2m+1) = 2^{2m+1} \mathbb{P}(A_r \geq 0 \ (1 \leq r \leq m), \ V_{m+1} = 0). \quad (67)$$

Replacing ≥ 0 by > 0 gives the corresponding identities for D .

[write] **5 October 2026.** The same encoding is Part I, Section 2, which adds the slack mixture (13). Here the endpoint is the composition condition at time $m+1$; there it is moved to time m by conditioning on the last gap.

The final step is not tested for area positivity. The last geometric variable is a slack variable, rather than an additional score in the half sequence. Imposing a positivity condition on A_{m+1} would change the count. For example, at $n = 2$ the valid weak half sequence (0) has $(A_1, V_2) = (0, -1)$ and $A_2 = -1$. At $n = 4$ the valid strong half sequence $(1, 1)$ has $A_1 = A_2 = 1$ and $A_3 = 0$. Both would be wrongly excluded by an extra test at the slack step.

14.2 Why the exponential factor is two

Before Landau inequalities are imposed, the number of half sequences is $\binom{M+m}{m}$. This is $\binom{2m-1}{m}$ for even length and $\binom{2m}{m}$ for odd length, each of order $2^n n^{-1/2}$. The additional integrated positivity constraint accounts for another $n^{-1/4}$ factor. This observation explains the exponent but is not itself a proof: survival and the terminal-velocity condition are dependent. The next two sections handle that dependence.

15 An elementary bounded bridge estimate

Let $T_j = X_1 + \dots + X_j$, with $T_0 = 0$ and the increment law in (64)–(65). Then

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 2, \quad \mathbb{E}e^{\theta|X|} < \infty \quad (0 < \theta < \log 2). \quad (68)$$

The support is $\{-1, 0, 1, \dots\}$, of lattice span one. All endpoint estimates in this section use these original integer coordinates.

Lemma 15.1. *There is a finite constant C such that*

$$\sup_{d \in \mathbb{Z}} \mathbb{P}(T_\ell = d) \leq C \ell^{-1/2} \quad (\ell \geq 1).$$

For each fixed $B > 0$, there are $c_B > 0$ and ℓ_B such that

$$\mathbb{P}(T_\ell = d) \geq c_B \ell^{-1/2} \quad \text{if } \ell \geq \ell_B, \ d \in \mathbb{Z}, \ |d| \leq B\sqrt{\ell}.$$

Proof. The sum of ℓ independent geometric variables has a negative-binomial law, so, for $d \geq -\ell$,

$$\mathbb{P}(T_\ell = d) = \binom{2\ell + d - 1}{\ell - 1} 2^{-2\ell - d}. \quad (69)$$

For $d < -\ell$ it is zero. Consecutive positive masses have ratio

$$\frac{\mathbb{P}(T_\ell = d+1)}{\mathbb{P}(T_\ell = d)} = \frac{2\ell + d}{2(\ell + d + 1)}.$$

Thus $d = -1$ is a mode (also $d = -2$ when $\ell \geq 2$), and

$$\sup_d \mathbb{P}(T_\ell = d) = \frac{1}{2} \binom{2\ell-2}{\ell-1} 4^{-(\ell-1)} = O(\ell^{-1/2}).$$

Stirling's formula, uniformly over $|d| \leq B\sqrt{\ell}$, gives

$$\mathbb{P}(T_\ell = d) = \frac{1}{\sqrt{4\pi\ell}} \exp\left(-\frac{d^2}{4\ell}\right) (1 + O_B(\ell^{-1/2})). \quad (70)$$

For clarity, the uniformity follows by writing $\binom{2\ell+d-1}{\ell-1} = \frac{\ell}{2\ell+d} \binom{2\ell+d}{\ell}$, applying Stirling to its three factorials, and expanding their logarithms about $d = 0$. In this window $d/\ell = O_B(\ell^{-1/2})$; the cubic remainder is $O_B(\ell^{-1/2})$, and the factorial error terms are $O(\ell^{-1})$. The Gaussian factor in (70) is at least $e^{-B^2/4}$, which proves the lower bound. $\square$

Lemma 15.2 (Bounded bridge). *For every fixed $B > 0$ there are $K > B$, $c > 0$, and ℓ_0 such that, for every $\ell \geq \ell_0$ and every integer $|d| \leq B\sqrt{\ell}$,*

$$\mathbb{P}\left(T_\ell = d, \max_{0 \leq j \leq \ell} |T_j| \leq K\sqrt{\ell}\right) \geq c\ell^{-1/2}. \quad (71)$$

Proof. Put $h = \lfloor \ell/2 \rfloor$ and $q = \ell - h$. For $\ell \geq 3$, both are comparable to ℓ . On a violation during the first h steps, condition on those steps, apply the uniform endpoint bound to the remaining q increments, and use Kolmogorov's maximal inequality. Since $\text{Var}(T_h) = 2h$, this gives

$$\begin{aligned} & \mathbb{P}\left(T_\ell = d, \max_{j \leq h} |T_j| > K\sqrt{\ell}\right) \\ & \leq \frac{C}{\sqrt{q}} \mathbb{P}\left(\max_{j \leq h} |T_j| > K\sqrt{\ell}\right) \leq \frac{C}{\sqrt{q}} \frac{2h}{K^2\ell}. \end{aligned}$$

For a violation at some $j > h$, the terminal condition $T_\ell = d$ implies

$$|T_\ell - T_j| \geq |T_j| - |d| > (K - B)\sqrt{\ell}.$$

The reversed suffix sums $R_i = X_\ell + X_{\ell-1} + \cdots + X_{\ell-i+1}$, $0 \leq i \leq q$, depend only on the last q increments and have the same maximal inequality. Condition on that block and use the atom bound on the first h increments:

$$\begin{aligned} & \mathbb{P}\left(T_\ell = d, \max_{h < j \leq \ell} |T_j| > K\sqrt{\ell}\right) \\ & \leq \frac{C}{\sqrt{h}} \frac{2q}{(K - B)^2\ell}. \end{aligned}$$

Hence, for a constant C_0 depending only on the increment law,

$$\mathbb{P}\left(T_\ell = d, \max_{j \leq \ell} |T_j| > K\sqrt{\ell}\right) \leq \frac{C_0}{\sqrt{\ell}} \left(\frac{1}{K^2} + \frac{1}{(K - B)^2} \right). \quad (72)$$

Choose K so large that the parenthesis times C_0 is at most $c_B/2$, where c_B is from Lemma 15.1. Subtracting (72) from the lower endpoint mass proves (71) with $c = c_B/2$. $\square$

The bridge is controlled by a deterministic velocity bound, so no conditioned bridge invariance principle or two-dimensional local limit theorem is needed.

[write] 5 October 2026. Part I, Lemma 5.1, proves the endpoint-weighted maximal upper bound $C\sqrt{\ell}/H^2$ by the same split; Lemma 15.2 is the lower-bound form used only in this Part.

16 The integrated walk with a fixed final velocity

For a starting state (a, v) , write

$$V_j = v + T_j, \quad I_j = a + \sum_{i=1}^j V_i, \quad \tau = \min\{j \geq 1 : I_j \leq 0\}.$$

Probabilities with this starting state are denoted by $\mathbb{P}_{(a,v)}$. The main probabilistic step is the following lemma.

Lemma 16.1. *For every fixed integer b ,*

$$\mathbb{P}_{(1,1)}(\tau > k, V_k = b) = \Theta(k^{-3/4}) \quad (k \rightarrow \infty). \quad (73)$$

The constants may depend on b . The applications below require only $b = 0, 1$.

16.1 The imported theorem and its hypotheses

Denisov and Wachtel [1, Theorem 1, equations (7)–(8), pp. 169–170] consider independent identically distributed increments Y satisfying

$$\mathbb{E}Y = 0, \quad \mathbb{E}Y^2 = 1, \quad \mathbb{E}|Y|^{2+\delta} < \infty \quad \text{for some } \delta > 0.$$

For the integrated process killed when its area ceases to be positive, their theorem supplies a positive harmonic function on

$$K_+ = \{z \in (0, \infty) \times \mathbb{R} : \mathbb{P}_z(Z_j \in (0, \infty)^2, \tau > j) > 0 \text{ for some } j \geq 0\}.$$

For a fixed starting point in K_+ , it gives a positive-constant survival asymptotic of order $k^{-1/4}$ and weak convergence of the conditional endpoint, with area divided by $k^{3/2}$ and velocity divided by $k^{1/2}$, to a probability measure μ whose density is displayed in that theorem.

Here $\sigma = \sqrt{2}$ and $Y = X/\sigma$. Equation (68) checks all the moment assumptions. Divide *both* area and velocity by σ , so the start becomes $(1/\sigma, 1/\sigma)$. This point belongs to K_+ already at time zero. Scaling by a positive number preserves the exit event. Consequently, for a constant $c_* > 0$,

$$\mathbb{P}_{(1,1)}(\tau > t) \sim c_* t^{-1/4}, \quad (74)$$

$$\mathcal{L}\left(\frac{I_t}{\sigma t^{3/2}}, \frac{V_t}{\sigma \sqrt{t}} \mid \tau > t\right) \Rightarrow \mu. \quad (75)$$

Theorem 1 has no lattice hypothesis. The normalized increments need not be integer valued; the exact lattice calculations of Section 15 remain in the unscaled coordinates. We use only Theorem 1, not any killed local-limit statement elsewhere in that paper.

16.2 Positive limiting mass at large area

We need more than survival alone: some surviving paths must have a large positive area and a bounded velocity at halfway time. The density formula in [1, Theorem 1] has the form

$$\bar{h}(a, b) = c_0 \int_0^\infty \int_0^1 w^{3/2} s^{-1/2} e^{-2w^2/s} q_{1-s}(a, -b; 0, -w) ds dw, \quad c_0 > 0, \quad (76)$$

where

$$q_u(x, y; z, v) = p_u(x, y; z, v) - p_u(x, y; z, -v)$$

and the Gaussian transition density is

$$p_u(x, y; z, v) = \frac{\sqrt{3}}{\pi u^2} \exp\left(-\frac{6(z-x-uy)^2}{u^3} + \frac{6(z-x-uy)(v-y)}{u^2} - \frac{2(v-y)^2}{u}\right). \quad (77)$$

Taking the ratio of the two Gaussian terms gives

$$\frac{p_u(a, -b; 0, -w)}{p_u(a, -b; 0, w)} = \exp\left(\frac{4w(3a-bu)}{u^2}\right). \quad (78)$$

If $a > 0$ and $b < 3a$, the right side exceeds one for all $0 < u < 1$, $w > 0$. Thus the integrand in (76) is positive and $\bar{h}(a, b) > 0$ in this region. In particular, every sufficiently high-area open rectangle with velocity close to zero has positive μ -mass.

[write] 5 October 2026, correction. The display (76) has the form of the density printed in [1, Theorem 1], which is misprinted: the true density of μ has $s^{-5/2}$ in place of $s^{-1/2}$ (and $c_0 = 2^{9/4}/\Gamma(1/4)$); see Remark 4.2 and Proposition 4.3. “The density formula ... has the form (76)” is therefore true of the printed display but not of μ . The argument of this subsection uses only the positivity of the outer weights and the ratio (78), both unchanged for the true density. Its conclusion, positive μ -mass of the rectangle, holds verbatim (Proposition 4.3(c): the integrand is positive where $a > 0$ and $b \leq 3a$).

Fix any sufficiently large L . In the normalized coordinates of (75), the open rectangle

$$(L/\sigma, (L+1)/\sigma) \times (-1/\sigma, 1/\sigma)$$

lies in that positive-density region. The Portmanteau theorem and (74) imply that, for some $c_L > 0$ and all large t ,

$$\mathbb{P}_{(1,1)}(\tau > t, Lt^{3/2} < I_t < (L+1)t^{3/2}, |V_t| < \sqrt{t}) \geq c_L t^{-1/4}. \quad (79)$$

The constant can be small, but it is fixed and strictly positive. No continuity of the density or local endpoint theorem is required for this use of weak convergence.

16.3 Proof of the endpoint lemma

Proof of Lemma 16.1. Put $t = \lfloor k/2 \rfloor$ and $\ell = k - t$. The upper bound follows by forgetting every area condition after time t . The Markov property, Lemma 15.1, and (74) yield

$$\begin{aligned} \mathbb{P}_{(1,1)}(\tau > k, V_k = b) &\leq \mathbb{P}_{(1,1)}(\tau > t) \sup_d \mathbb{P}(T_\ell = d) \\ &= O(t^{-1/4} \ell^{-1/2}) = O(k^{-3/4}). \end{aligned}$$

For the lower bound, take $B = 2$ in Lemma 15.2, and fix its constant K . Then choose $L = 4(K+1)$ in (79). Given a halfway state (a, v) in that event, $|v| < \sqrt{t} \leq \sqrt{\ell}$. Since b is fixed, $|b - v| \leq 2\sqrt{\ell}$ for all large k . The independent remaining increments therefore have probability at least $c/\sqrt{\ell}$ of satisfying

$$T_\ell = b - v, \quad \max_{j \leq \ell} |T_j| \leq K\sqrt{\ell}.$$

During this event every remaining velocity is at least $-(K+1)\sqrt{\ell}$. For every $j \leq \ell$, the area obeys

$$I_{t+j} = a + jv + \sum_{i=1}^j T_i \geq a - (K+1)\ell^{3/2} > Lt^{3/2} - (K+1)\ell^{3/2} > 0.$$

The last inequality holds for all sufficiently large t , since $\ell \in \{t, t+1\}$ and $L = 4(K+1)$. Thus the entire final block survives, and $V_k = b$. Multiplying the conditional bridge bound by (79) gives

$$\mathbb{P}_{(1,1)}(\tau > k, V_k = b) \geq c't^{-1/4}\ell^{-1/2} \geq c''k^{-3/4}.$$

This proves both bounds. $\square$

[write] **5 October 2026.** Part I, Theorem 6.1, proves the equivalent $\mathbb{P}_z(\tau > k, V_k = b) \sim c_z f_\star(0) \sigma^{-1} k^{-3/4}$ for every positive integer start; this proof is a second route to the order of magnitude, from the start $(1, 1)$.

17 Application to the two counting sequences

17.1 Weak upper bound and the extra endpoint step

In Proposition 14.1, put $I_r = A_r + 1$. Since the areas are integers, $A_r \geq 0$ if and only if $I_r > 0$. The shifted walk has starting state $(I_0, V_0) = (1, 1)$, so (74) applies.

Set $t = \lfloor m/2 \rfloor$, and let $b = -1$ for even length or $b = 0$ for odd length. Dropping all area constraints after t and retaining the endpoint at time $m+1$ gives

$$\begin{aligned} \mathbb{P}(A_r \geq 0 \ (1 \leq r \leq m), \ V_{m+1} = b) \\ \leq \mathbb{P}_{(1,1)}(\tau > t) \sup_d \mathbb{P}(T_{m+1-t} = d) = O(m^{-3/4}). \end{aligned}$$

There is no new area condition at time $m+1$ in this bound. Multiplication by 2^n gives $C(n) = O(2^n n^{-3/4})$, and $D(n) \leq C(n)$ gives the same upper bound for the strong count.

17.2 Strong lower bound from two fixed gaps

For the strong event, force

$$G_1 = 1, \quad G_{m+1} = 0. \tag{80}$$

These independent requirements cost exactly $(1/4)(1/2) = 1/8$. The first step gives $(A_1, V_1) = (1, 1)$, so strict positivity is already satisfied. The next $m-1$ tested steps are an integrated walk from this strictly interior starting state. This avoids applying the imported theorem to the boundary starting point $(0, 1)$.

The final slack increment is $X_{m+1} = -1$. Therefore the tested-time endpoint needed at time m is

$$V_m = \begin{cases} 0, & n = 2m, \\ 1, & n = 2m + 1. \end{cases}$$

Let b denote the indicated tested-time target. Independence and Lemma 16.1 show that the event in the strong counting identity has probability at least

$$\frac{1}{8} \mathbb{P}_{(1,1)}(\tau > m-1, V_{m-1} = b) = \Omega(m^{-3/4}).$$

Here the time index $m-1$ belongs to the walk restarted after the forced first step. The endpoint velocity is still the absolute velocity, not a displacement. The final forced slack restores the required original terminal velocity -1 or 0 , respectively.

Thus $D(n) = \Omega(2^n n^{-3/4})$. Since $C(n) \geq D(n)$, the same lower bound holds for C . This completes the growth part of Theorem 13.1, with constants valid across both parities.

[write] **5 October 2026.** Part I, Section 9, decomposes the strong count over the first gap $g \geq 1$, (49), instead of forcing $G_1 = 1$. It obtains the strong amplitude, and comparison with $D_n \sim rC_n$ gives the harmonic identity (51). The forced-gap class here keeps only a subset of the paths, as Section 19 says.

18 Monotonicity and threshold inversion

Proposition 18.1. *$C(n)$ is nondecreasing for all $n \geq 0$. $D(n)$ is nondecreasing for $n \geq 2$. The initial exception is $D(1) = 1 > D(2) = 0$.*

Proof. From length $2m$ to length $2m+1$, keep the same half sequence. The cap increases from $m-1$ to m , while all tested excesses remain unchanged. Reflection recovers the original half sequence, so the map is injective and preserves weak or strict inequalities.

From length $2m+1$ to length $2m+2$, send

$$(s_1, \dots, s_m) \mapsto (s_1, \dots, s_m, m).$$

The new cap is m . Earlier excesses are unchanged, and the new one is

$$A_{m+1} = A_m + m - m = A_m.$$

For $m \geq 1$ this preserves both weak and strict inequalities; removing the last half score proves injectivity. For $m = 0$ it preserves only the weak condition, because $A_0 = 0$. Directly checking $n = 0, 1, 2$ gives the asserted initial behavior. $\square$

[write] **5 October 2026.** Not in Part I, which has only eventual strict increase of both counts (Section 10). Together: C_n is nondecreasing for all n and D_n for $n \geq 2$, both eventually strictly increasing; the table of Section 20 shows equalities $C(1) = C(2)$ and $C(3) = C(4) = 2$, $D(3) = D(4)$, $D(5) = D(6)$.

Proof of the inverse formula. Fix $E = C$ or $E = D$. The growth theorem makes $N_E(x)$ finite for all large x and ensures $N_E(x) \rightarrow \infty$. Write $n = N_E(x)$ and $y = \log_2 x$. Minimality gives $E(n-1) < x \leq E(n)$. For sufficiently large x , both $n-1$ and n lie beyond the finite exceptions in the two-sided growth bounds, so

$$c 2^{n-1} (n-1)^{-3/4} < x \leq C 2^n n^{-3/4}$$

for fixed positive c, C . Taking logarithms yields

$$y = n - \frac{3}{4} \log_2 n + O(1). \tag{81}$$

First $n \asymp y$ and then $n = y + O(\log y)$. Hence $\log_2 n = \log_2 y + O((\log y)/y)$. Substituting in (81) proves (59). The isolated decrease in D is irrelevant: minimality, eventual bounds, and unboundedness suffice for this argument. $\square$

[write] **5 October 2026.** Part I, Proposition 10.1, sharpens the $O(1)$ for the smooth inverse to the constant $-\log_2 A_E$ plus $o(1)$ and gives vanishing-width ceiling brackets for N_E ; Corollary 10.2 adds the eventual zero-or-one gap between N_C and N_D . Its proof also shows that a global $o(1)$ error for N_E is false. This is consistent with Section 19: a Θ estimate alone gives no constant, and the constant comes from the equivalents.

19 What is proved and what remains open

The proof establishes two-sided growth bounds, with positive finite constants, and an inverse accurate to a bounded additive error. It also gives an integrated-walk endpoint estimate valid for every fixed integer terminal velocity from the interior start $(1, 1)$.

It does not identify a limit for $C(n)/(2^n n^{-3/4})$ or $D(n)/(2^n n^{-3/4})$, nor prove that such limits exist on either parity. The two-sided bounds imply that $D(n)/C(n)$ stays bounded away from zero for all large n , but they do not prove convergence of this ratio or the numerical proportion suggested by finite data. An $O(1)$ error in the inverse cannot be sharpened to a fixed constant from a Θ estimate alone.

Three natural next steps are:

1. **Endpoint-conditioned density estimates.** Establish a local estimate for a fixed terminal velocity, with integrated area ranging over its natural $n^{3/2}$ scale, and prove tail bounds strong enough to sum it. Conditional weak convergence alone does not determine the probability of a single velocity value.
2. **Leading constants and parity.** Use such an estimate, together with the exact initial and final gap constraints, to determine whether the even and odd normalized counts converge and whether their limits agree. The fixed-gap lower bound used here intentionally retains only a subset of valid paths, so it cannot recover a leading constant.
3. **Strong-to-all ratios and corrections.** Analyze the weak and strict boundary conditions sharply enough to compare their amplitudes, then justify any lower-order powers, logarithmic terms, or other correction expansions. No transseries or full correction expansion is established in this report.

No inference about differential finiteness or analytic continuation is drawn from the order-of-growth estimates. Such conclusions require additional arguments, not numerical fitting of the exponent.

[write] **5 October 2026.** Status after Part I. Steps 1 and 2 are answered there for the purposes stated: Theorem 6.1 gives the fixed-velocity endpoint equivalent (without localizing the area), and Section 7 shows that both parities have one limit with no parity multiplier. The strong-to-all limit of step 3 is $e^{-\lambda}$ (Section 8). Correction terms, logarithmic terms and transseries remain open in both Parts. The non-claims above remain true of this Part as printed. See Section 21.

20 Exact checks and reproducibility

The accompanying program is a finite consistency check of the combinatorial encoding and its indexing. It is not a numerical proof of an asymptotic claim, and the probabilistic lemmas above do not depend on the program's output.

20.1 Independent finite formulations

The checker uses four representations:

1. A half-score dynamic program on the last score and the running score sum, extending nondecreasing half sequences directly.
2. A composition dynamic program on the used gap total and weighted gap sum, appending the

uniquely determined final slack without another area test.

3. An integrated-walk dynamic program on velocity and area, using endpoint reachability to bound transitions and including the final untested slack step.
4. Brute enumeration of full nondecreasing score lists, imposing reflection symmetry and every proper Landau inequality in the full list.

The first three agree for both weak and strict inequalities at every $n = 0, \dots, 38$ (156 pairwise comparisons); the full enumeration agrees for $n = 0, \dots, 9$ (20 comparisons). The computed prefixes match 35 reference terms of A345470 and 39 of A351869, independently checked against the OEIS entry pages on 2 October 2026, starting at $n = 0$.

Exhaustive tests for $n = 2, \dots, 17$ compare 26,364 half candidates with 26,364 independently generated compositions. They check reflection of excesses, the gap bijection, equal geometric mass, 22,987 valid monotonicity injections, the fixed-gap lower-bound class, and both final-slack counterexamples. Exact rational and symbolic checks also cover the negative-binomial identities, increment moments, Gaussian density-ratio algebra, and finite threshold bookkeeping. These checks support the formulas and indexing; they do not certify the probability limit theorem or replace any proof above.

[write] **5 October 2026.** The package is shipped under other names (`checks/reader_validation.md` is 114-growth-checks-reader_validation.md, the scripts are in `code/` and the records in `data/`; README, “Delivered path to shipped path”). The 35 and 39 reference terms are OEIS data (CC BY-SA 4.0). The “Gaussian density-ratio algebra” checked here is the ratio (78), which is the same for the printed and the true density.

The first values are included for orientation; no extrapolation is used.

n	$C(n)$	$D(n)$	n	$C(n)$	$D(n)$
0	1	1	9	48	30
1	1	1	10	64	39
2	1	0	11	161	103
3	2	1	12	223	141
4	2	1	13	557	363
5	5	3	14	796	514
6	6	3	15	1971	1301
7	15	9	16	2887	1894
8	19	11	17	7090	4727

20.2 Replay contract

The reader archive contains this TeX source and PDF, a standard-library Python checker, fixed reference prefixes with provenance, generated exact counts and check summary, a deterministic PDF build script, a replay driver, and a strict SHA-256 manifest. No downloaded source PDFs are bundled. Bibliographic links below identify the primary sources.

Run `python3 replay.py` from the extracted package directory. The replay first validates the exact manifest path set, file types, sizes, and hashes. It runs the finite checker both normally and under `python3 -O` and compares both outputs with the archived results. The mathematical and schema mutation campaign rejects 37 named mutants in 74 normal and optimized runs, each with its prescribed diagnostic rather than a syntax error or crash. The suite also rejects 17 damaged,

absent, unexpected, or unsafe manifest and package variants, with the same protections active under optimization. All requirements use explicit exceptions rather than Python `assert`, so optimized execution cannot silently disable checks.

If `pdflatex` is installed, `python3 replay.py --build-pdf` also builds the PDF twice in clean temporary directories and compares the bytes with each other and with the archived PDF. Byte-for-byte reproduction requires the same TeX engine, packages, and fonts. The recorded toolchain is included in `build-environment.txt`; a different toolchain may typeset an equivalent PDF with different bytes. Arithmetic verification uses only Python’s standard library and does not require network access or external source downloads.

21 Further questions and research

[write] (5 October 2026.) This section was added in the write, under Vladimir’s standing rule of 4 October 2026. It collects every claim of Part II stated without proof or left open, and records how Part I has re-scoped it.

1. *Endpoint-conditioned density estimates* (Section 19, step 1). Its purpose, the probability of a single terminal velocity with its constant, is achieved by Part I, Theorem 6.1, by smoothing over a short final block instead of a local estimate in the area. A local estimate with the area on its $n^{3/2}$ scale and summable tails, the step as stated, is not proved in either Part; [1, (12)] states a conditional local limit theorem that Part I deliberately does not use.
2. *Leading constants and parity* (step 2). Answered by Part I: one amplitude A for both parities, with $A = \sqrt{6} \Gamma(3/4) \pi^{-3/2} V_{\text{DW}}(1/\sqrt{2}, 1/\sqrt{2})$ after the correction of Section 4.2. Open: the numerical value (Section 12.1, item 1).
3. *Strong-to-all ratios and corrections* (step 3). The ratio is answered, $D(n)/C(n) \rightarrow e^{-\lambda}$. Lower-order powers, logarithmic terms and correction expansions remain open (Section 12.1, items 2 and 5).
4. *Differential finiteness and analytic continuation*. This Part draws no inference from the growth estimates, and neither does Part I. Whether $C(z)$ or $D(z)$ is D-finite is not addressed in this report.

Refuted (Remark 4.2): “The density formula in [1, Theorem 1] has the form (76)” is true of the printed display, but that display is not the density of μ : the true density has $s^{-5/2}$. No conclusion of this Part changes. Nothing else in this Part failed a check: the encoding, the reflection identity, the 2^{-n} mass for both parities, the forced-gap time indices and the monotonicity injections were rechecked at the placement.

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- [1] D. Denisov and V. Wachtel, *Exit times for integrated random walks*, Annales de l’Institut Henri Poincaré, Probabilités et Statistiques **51** (2015), no. 1, 167–193. Part I: Theorem 1, equations (7)–(8), and its displayed density on pp. 169–170. Part II: Theorem 1, equations (7)–(8), and the density immediately following (8), pp. 169–170. **[write]** The displayed density is misprinted (Remark 4.2); Remark 2 (p. 170) and Lemma 16 (p. 186) identify the correct one. The misprint is in this journal version only; arXiv:1207.2270v1 (2012) states Theorem 1 with a different form of the density (Remark 4.5; added 5 October 2026). doi:10.1214/13-AIHP577. [Open full text](#); <https://www.numdam.org/articles/10.1214/13-AIHP577/>
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